Response to the Editor and Reviewer Comments

2025-RA-20071.R2

Humans' Fast and Slow Responses to Competition from Internal Generative AI in Online Q&A Communities

We sincerely thank the Senior Editor (SE), the Associate Editor (AE), and the reviewers for the opportunity to revise our paper. The revision has benefited greatly from the detailed guidance provided by the review panel. Through email exchanges with the SE and a subsequent face-to-face meeting, we gained a clearer understanding of the key issues to address and the most promising direction for the revision. We are grateful for the SE's guidance, and the revised manuscript has been developed in accordance with that guidance as well as the constructive comments from the AE and reviewers.

Based on this guidance, we have substantially revised the paper in the following areas:

- **(a) Reframing of the paper.** Following the SE's suggestion, we examine human contributions in the presence of an AI answer through a *competition* lens. Consistent with this revised framing, and to sharpen the focus of the paper, we have streamlined the storyline around human answers and removed the analysis of viewers.
- **(b) Development of an overarching theoretical framework.** Following the AE's and reviewers' suggestions, we have developed an overarching theoretical framework. Drawing on competitive dynamics theory (Chen et al., 2007; Chen, 2009) and Kahneman's two systems of thinking (Evans & Stanovich, 2013; Kahneman, 2003, 2011), we explain how a human appraises an AI answer, decides whether to contribute, and, if so, determines the quality of the answer to provide.
- **(c) Refinement of the focal AI-answer features.** Building on and extending our prior analyses, we argue that the effect of an AI answer depends on a profile of features: label, length, and quality. We use this profile to theorize the effect of internal GenAI in our empirical setting. In line with the AE's and reviewers' suggestions, we have removed the position effect from the study.
- **(d) Clarification of the paper's positioning and contribution.** Per the SE's suggestion, we have sharpened the paper's positioning relative to prior work, differentiated our paper from related studies, and highlighted the uniqueness and contribution of our study.

We have also carefully addressed all additional concerns raised by the reviewers. Below, we present our detailed, itemized responses to each comment. We hope the review panel finds the paper significantly improved from the previous version.

References:

Chen, M.-J. (2009). Competitive dynamics research: An insider's odyssey. *Asia Pacific Journal of Management*, 26(1), 5–25.

Chen, M.-J., Su, K.-H., & Tsai, W. (2007). Competitive tension: The awareness–motivation–

capability perspective. *Academy of Management Journal*, 50(1), 101–118.

Evans, J. St. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, 8(3), 223–241.

Kahneman, D. (2003). A perspective on judgment and choice: Mapping bounded rationality. *American Psychologist*, 58(9), 697–720.

Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.

Su, Y., Zhang, K., Wang, Q., & Qiu, L. (2025). Generative AI and human knowledge sharing: Evidence from a natural experiment [Working paper]. SSRN.

SE Comments

SE-0

To evaluate your work, I worked with the same Associate Editor and review panel to evaulate your work. As you are well-aware, this panel is familiar with both your topic and your empirical approach. I would like to begin by expressing my appreciation for the care and effort the Associate Editor and the reviewers invested in evaluating this revision. The Associate Editor's report in particular reflects a careful reading of the manuscript, the revision document, and the reviewers' comments, as well as thoughtful reflection on how best to proceed.

I also want to acknowledge the Associate Editor's reluctance to recommend that the manuscript move forward in the review process. Their report reflects a sincere effort to balance the promise of the research with the review team's concerns. I appreciate that deliberation and the care taken in reaching their recommendation.

The review team's assessment remains mixed, with most reviewers continuing to express significant concerns about the manuscript. However, after carefully reviewing the revision, the Associate Editor's report, and the reviewers' comments, I found myself reluctant to reject the paper. The author team has addressed many of the panel's empirical concerns. However, they have not made sufficient progress in addressing the panel's theoretical concerns about the paper. Given the progress on one dimension, but not the other, this presents a quandary - and the more prudent editorial path is likely to reject the paper.

However, I believe there is a narrow path forward for the paper if the theoretical issues identified in the review package can be addressed convincingly.

Accordingly, I am willing to offer one additional revision opportunity.

I want to be clear: this revision must do more than resolve empirical concerns; it must squarely address the theoretical issues raised by the Associate Editor and the reviewers. While the empirical work in the paper is promising, the manuscript cannot move forward in the review process unless the theoretical explanation of the observed effects becomes substantially clearer and more convincing.

<u>Response</u>

We sincerely thank the review panel for the recognition of our progress on the empirical dimensions and the opportunity to revise the paper. We are fully committed to addressing these theoretical concerns substantively and demonstrating a convincing explanation of the mechanisms underlying our findings in this revision.

SE-1

The Associate Editor has done an excellent job summarizing the review team's concerns. Rather than repeating their detailed discussion, I will highlight the issues I believe are essential to address in the revision.

First, the paper must establish a clearer and stronger theoretical foundation.

The central issue identified by the Associate Editor and all three reviewers is that the *manuscript currently documents empirical patterns without providing a sufficiently convincing theoretical explanation of what mechanisms are generating those patterns.*

As the Associate Editor notes, the paper reasonably demonstrates that the treatment effects are robust and that the empirical patterns are unlikely to be artifacts of model specification or data issues. However, documenting the pattern alone is not sufficient. The review team needs to be convinced that the manuscript provides a plausible and theoretically grounded explanation of the mechanisms underlying those patterns.

Currently, the manuscript relies on three proposed mechanisms: content effect, source effect, and position effect. Reviewer 2 questions whether the selection of these mechanisms is theoretically justified relative to other mechanisms discussed in the literature. Reviewer 3 similarly notes that the three mechanisms appear to be introduced largely in isolation rather than as part of a coherent theoretical argument. The Associate Editor raises a related concern: the manuscript lacks an overarching theoretical framework explaining why these mechanisms should operate together.

In the next revision, the theoretical argument must be reconstructed to present a coherent explanation of how internal generative AI changes participation incentives and evaluation processes in online knowledge communities.

This requires more than refining the discussion of the three effects. The paper must articulate:

• the underlying theoretical problem the study addresses

• the mechanism or mechanisms through which internal GenAI alters contributor behavior

• how the proposed mechanisms follow from existing theory

Put simply, the paper must move from describing empirical regularities to explaining the behavioral processes that generate those regularities.

Please note that, because MISQ prioritizes theoretical contributions, this concern must be substantively addressed in this revision. I view this as the "paper-breaker" - if you can't address the concern about the theoretical foundation, then the paper can not advance to another round of review.

Response

We have addressed this concern as a central focus of the revision. Following your suggestion in Comment SE-4, we have developed a new theoretical framework centered on the competitive encounter between human answerers and AI answers generated by the platform's internal GenAI system. Specifically, we theorize how the AI answer affects two decisions made by human answerers: the **entry decision**, or whether to contribute an answer at all, and the **crafting decision**, which concerns how good the human answer will be and how it will be produced, if the answerer decides to contribute.

We adopt a competition lens informed by competitive dynamics theory (Chen et al., 2007; Chen, 2009). This lens matches the setting because the AI answer sits in the same thread and on the same screen as any human answer that follows. A later human answer must therefore be worth reading in the presence of the AI answer. It also gives us richer predictions than a simple substitution account, which would predict only that AI answers reduce human contributions. By contrast, the competition lens explains why an AI answer may discourage human entry in some cases while eliciting higher-quality human answers from those who do enter.

The revised framework treats the AI answer as a profile of three signals:

(a) **AI label.** The label identifies the source of the existing answer and tells the answerer what kind of competitor this is. Because machine authorship is apparent at a glance, the label can shape the answerer's initial appraisal before the AI answer's content is read. In our theory, the label leads answerers in this technical Q&A setting to appraise the AI as a weak competitor, which strengthens the incentive to enter.

(b) **AI-answer length.** Length is also available at a glance. It signals how much of the answerable ground the AI answer appears to have taken. A longer AI answer can appear to cover more possible causes or fixes, which reduces the perceived room left for a human contribution and weakens the incentive to enter.

(c) **AI-answer quality.** Quality is not available at a glance. It becomes available only when the answerer reads the AI answer and evaluates whether its reasoning, diagnosis, or code fits the specific problem. Once a human answerer has decided to contribute, a higher-quality AI answer serves as a better foundation to build on, either by providing adaptable material such as a correct diagnosis or working code that can be reused, or by charting a viable direction that the answerer can develop into a full answer.

We link these signals to the two decisions through the two systems of thinking theory (Evans & Stanovich, 2013; Kahneman, 2003, 2011). The AI label and AI-answer length are **glance signals** processed through fast, low-effort System 1 thinking, whereas AI-answer quality is a **reading signal** processed through deliberate System 2 thinking. Similarly, entry is a System 1 decision because it can be settled quickly before close reading; the crafting decision is a System 2 decision because it is made after the answerer has committed to answering and determines both how good the submitted answer will be and how it will be produced.

This classification yields a **same-level principle**: a signal affects a decision when both occupy the same level of processing, while cross-level effects are expected to be null. Thus, the AI label and AI-answer length shape entry, but AI-answer quality does not. Conversely, AI-answer quality shapes human answer quality by serving as a foundation the answerer can reuse and redirect to lower the cost of producing a high-quality answer, but the AI label and AI-answer length do not systematically shape human answer quality once the answerer has entered.

In line with this framework, the revised manuscript develops six hypotheses (see Figure SE-1 below): the AI label increases human answer supply; the effect of an AI answer on human answer supply becomes more negative as AI-answer length increases; AI-answer quality does not appreciably shape human answer supply; the AI label does not affect human answer quality; AI-answer length does not affect human answer quality; and the effect of an AI answer on human answer quality becomes more positive as AI-answer quality increases.

We then test these predictions using our empirical data. In light of this revised conceptual framework, we have removed the analysis of viewer recognition and the position effect to keep the paper theoretically focused and internally coherent.

We hope the review panel finds that the revised framework substantially strengthens the theoretical foundation of the paper and satisfactorily addresses your concern.

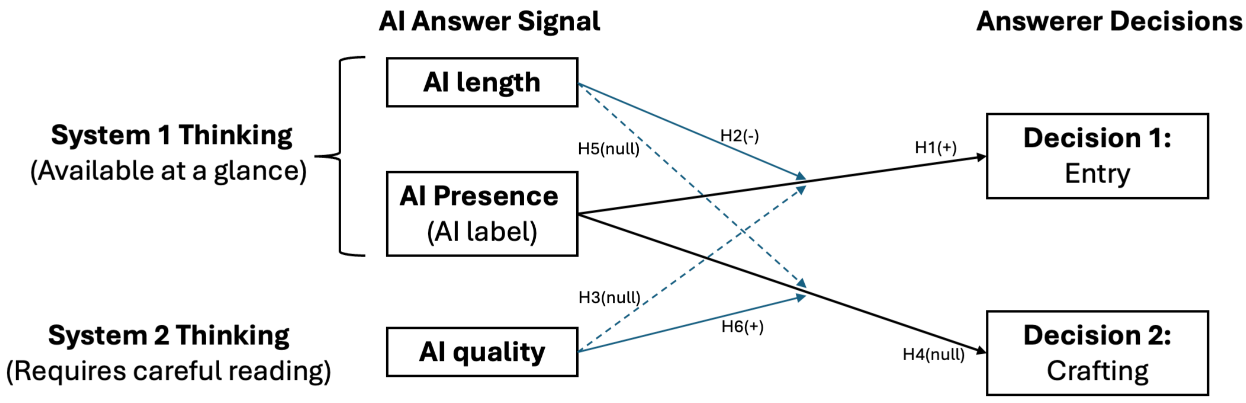

Figure SE-1. Conceptual framework

References:

Chen, M.-J. (2009). Competitive dynamics research: An insider's odyssey. *Asia Pacific Journal of Management*, 26(1), 5–25.

Chen, M.-J., Su, K.-H., & Tsai, W. (2007). Competitive tension: The awareness–motivation–capability perspective. *Academy of Management Journal*, 50(1), 101–118.

Evans, J. St. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, 8(3), 223–241.

Kahneman, D. (2003). A perspective on judgment and choice: Mapping bounded rationality. *American Psychologist*, 58(9), 697–720.

Kahneman, D. (2011). Thinking, fast and slow. Farrar, Straus and Giroux.

SE-2

Second, the empirical tests of the mechanisms must more directly support the theoretical explanation.

Many of the reviewers' concerns stem from the fact that the empirical analyses intended to test the proposed mechanisms do not provide clear evidence for them.

For example:

Reviewer 1 and Reviewer 3 both question whether the analysis in Section 6.3.1.1 provides convincing evidence for the proposed content effect, noting that answer length alone is not a reliable measure of answer quality.

Reviewer 2 and Reviewer 3 raise concerns about the operationalization of the position effect, particularly the use of the "first-position seeker" construct and its connection to the underlying theoretical logic.

All three reviewers question the inference of the source effect, noting that attributing residual treatment effects to an AI authority mechanism is not justified unless other *plausible mechanisms have been ruled out.*

These issues suggest that the current analyses of the mechanism do not yet provide convincing empirical support for the theoretical claims. In the revision, the mechanism tests should be reconsidered to more directly correspond to the theoretical arguments being advanced.

Please note, that this may require your team to develop a novel theoretical explanation. Given you results counter Su et al. (2023), you may need to develop a novel theoretical account for your findings.

Response

We are grateful for this guidance. Following your suggestion, we have revised both the theoretical framework and the empirical analysis so that the tests correspond more directly to our theory.

First, we now explicitly incorporate both AI-answer length and AI-answer quality into our framework and examine their distinct effects. We theorize AI-answer length as a signal of answer coverage, which captures how much of the answerable ground the AI answer appears to have taken. We measure AI-answer quality using LLM-based evaluations of content quality. This distinction allows us to test whether length and quality affect different human decisions in different ways.

Second, we reconsidered the position-effect analysis in light of the reviewers' concerns. A direct test of this effect would require data on how answerers perceive and value answer position, which our current dataset does not provide. Following your guidance and the revised conceptual framework, we have removed the position-effect analysis from the manuscript.

Third, we replaced the residual analysis with a direct assessment of the AI-label effect. We agree that attributing residual treatment effects to an AI authority mechanism was not sufficiently justified without ruling out other plausible explanations. In the revised manuscript, the AI label is theorized and tested as a glance signal that shapes the entry decision.

Taken together, these revisions ensure our empirical analyses are more specifically targeted at the theoretical argument we intended to test. The revised account is organized around the fast and slow processing of AI-answer signals: the AI label and AI-answer length are processed as glance signals that shape entry, while AI-answer quality is processed through more deliberate evaluation and shapes the quality of subsequent human answers.

This revised framework also helps reconcile our findings with Su et al. (2025). Please see our response to SE-3 below. We also provide a detailed discussion on page 48 of the revised manuscript.

Reference:

Su, Y., Zhang, K., Wang, Q., & Qiu, L. (2025). Generative AI and human knowledge sharing: Evidence from a natural experiment [Working paper]. SSRN.

SE-3

Third, the manuscript must more clearly situate its findings relative to prior work.

Several reviewers raise concerns about the paper's positioning relative to existing literature, particularly Su et al. (2023).

As Reviewer 1 notes, the current explanation for the divergence between your findings and prior work is not yet convincing. The revision should clarify why the mechanisms proposed in your paper would lead to different outcomes in your setting and what theoretical insight emerges from this comparison.

This clarification is important for establishing the paper's contribution.

Response

Following the suggestion of the review panel, we have further differentiated our study from Su et al. (2025) and, more broadly, from the two other closely related studies. Table SE-3 positions the four studies along the dimensions that drive our identification.

Table SE-3. Positioning relative to the three most closely related studies

Dimension	Xue et al. (2026)	Shan & Qiu (2026)	Su et al. (2025)	Our study
Primary outcome	Question (supply & quality)	Answer (supply, length & readability)	Answer (supply & length)	Answer (supply & quality)
Domain and basis of contribution value	Technical; correctness	Technical; correctness	General-interest; experience and interpretation	Technical; correctness
Locus of AI	External	External	Platform-internal	Platform-internal
Timing of AI answer	NA	NA	years after posting (treated-group mean pre-policy tenure ≈ 49 months)	immediately after posting
AI's relation to the human answerer	None – diverts questions pre-arrival	Collaborator	Supplementary contributor	Direct competitor
AI identifiable	No	No	Yes	Yes
Separates AI *label* from AI *content*	No	No	Not the focus	Yes
Research design	Natural experiment	Natural experiment	Natural experiment and lab experiment	Randomized field experiment

Our study focuses on competition between internal GenAI and human answerers, whereas the two external GenAI studies examine a different form of competition. Xue et al. (2026) study competition with the platform, where external GenAI diverts questions before they reach the community. Shan & Qiu (2026) study the invisible facilitation of human answers, where external GenAI helps humans craft their responses. Four features make our setting distinctive. First, the platform deploys the GenAI under direct governance, so its presence is a design choice rather than an uncontrollable environmental shift, which places the findings within managerial reach. Second, the GenAI responds almost immediately, before any human contributor arrives, so human contributors face the most competitively demanding configuration. Third, the AI responses carry an explicit label, so we estimate the effect of the AI label separately from content attributes such as length and quality. Fourth, we randomize the assignment of AI answers at the question level, which permits clean causal inference and lets us compare questions whose earliest answer is AI-generated with otherwise identical questions whose earliest answer is human-generated.

Our study also differs from Su et al. (2025) in domain, and this difference reshapes the mechanism. Su et al. (2025) study Quora, a general-interest platform whose content is experiential and open-ended. SegmentFault is a technical platform, and each question calls for a correct and efficient solution. This difference changes how contributors read the length of an AI answer. In our technical setting, contributors treat length as a fast surface cue for how much of the question the AI has already covered, so a question that looks well covered suppresses entry. On Quora, many distinct answers coexist and earn value at once, and human value rests on personal experience and perspective. When contributors do not read length as a coverage cue, the deterrent channel stays muted, and the positive AI label effect prevails.

Our study also differs from Su et al. (2025) in timing. In Su et al. (2025), the AI answer is introduced long after the original question was posted (average pre-policy tenure is roughly 49 months), by which point at least one human answer is already present (Su et al. 2025, pp. 15, 22). The AI therefore enters as a retrospective supplement to an established discussion. In our setting, the AI answer is generated quickly after the question posting and before human contributors arrive, so it is the rival that a contributor evaluates at the moment of entry.

Our framework accommodates both sets of results. The effect of the AI label alone on supply is positive and significant (0.188, $p<0.05$), which aligns with Su et al. (2025). The interaction with AI-answer length is negative (-0.046, $p<0.001$), which turns the average total effect negative. Contributors read a long AI answer as broad coverage and stay out. The AI label effect dominates in Su et al. (2025) because the deterrent role of length does not materialize on a general-interest platform, whereas both channels operate in our technical setting. Whether an identified AI contributor complements or displaces human contribution is therefore not a fixed property of internal GenAI. It is a joint function of the AI label effect and the AI content effect, and it depends on the domain, on how much content the AI provides, on how readily contributors read that content as coverage, and on when the AI answer arrives relative to human contributors.

References:

Shan, G., & Qiu, L. (2026). Examining the Impact of Generative AI on Users' Voluntary Knowledge Contribution: Evidence from a Natural Experiment on Stack Overflow. *Information Systems Research*, 37(2), 1021–1041.

Su, Y., Zhang, K., Wang, Q., & Qiu, L. (2025). Generative AI and human knowledge sharing: Evidence from a natural experiment [Working paper]. SSRN.

Xue, J., Wang, L., Zheng, J., Li, Y., & Tan, Y. (2026). Can ChatGPT Kill User-Generated Q&A Platforms? *Information Systems Research*.

SE-4

Fourth, the manuscript should clarify the relationship between theory and empirical *findings.*

Reviewer 2 raises a concern that the theoretical development may have been shaped after the empirical results were observed. Whether or not that characterization is accurate, the manuscript should clarify the logical ordering between theory and empirical analysis.

In particular, the revision should explicitly acknowledge alternative theoretical predictions that exist in the literature. Some prior work suggests that AI-generated answers could increase rather than decrease human participation. Addressing these alternative possibilities directly will strengthen the credibility of the theoretical argument.

So what to do? Given the strength and breadth of objections to your work?

The review package suggests several directions to strengthen the paper's theoretical contribution. While the choice of approach ultimately rests with the authors, the following directions strike me as particularly promising.

One possible direction is to frame the paper more explicitly around participation incentives in online knowledge communities. In this framing, the introduction of internal GenAI could be theorized as altering the perceived marginal value of human contributions and the expected rewards associated with contributing.

A second possibility is to develop a stronger theoretical account of authority and credibility signals in human–AI interaction. Several reviewers note that the paper currently assumes that users attribute authority to AI-generated answers, but this assumption requires stronger theoretical justification in the context of generative AI systems.

A third direction is to frame the study more broadly as examining competition and complementarity between AI-generated and human-generated knowledge in digital communities. Such a framing could help explain when AI discourages human participation and when it might instead stimulate additional contributions.

These suggestions are not intended to prescribe a specific solution. Rather, they illustrate the type of theoretical narrative that will be necessary for the paper to make a convincing contribution.

Response

We thank you for laying out these possible directions for strengthening the manuscript. After carefully considering them, we have reorganized the paper around the third direction. The revised manuscript frames the introduction of internal GenAI as creating a competitive encounter between an AI answer and potential human answerers. We then rebuild the theoretical arguments around this framework, and the updated theoretical framework aligns more closely with both the literature and the phenomenon of interest.

The revised theoretical development now begins by casting the human-AI encounter as a competitive episode. As noted in our response to Comment SE-3, this framing also helps explain the difference between our findings and those of Su et al. (2025). In our technical Q&A setting, the AI answer appears early and can be evaluated by potential human answerers at the moment they decide whether to contribute. Human answerers therefore appraise the AI answer as a potential competitor using observable features, consistent with competitive dynamics theory

(Chen et al., 2007; Chen, 2009). In particular, they may rely on the AI label, AI-answer length, and AI-answer quality when making entry and crafting decisions. This provides the core theoretical contribution of the revised manuscript.

We also explicitly acknowledge alternative predictions in the literature. Prior work suggests that AI-generated answers may stimulate, rather than reduce, human participation. Our revised framework accommodates this possibility by treating the AI answer as a profile of multiple signals rather than as a uniform treatment. Some features may encourage contribution, while others may deter it.

This framework clarifies the relationship between theory and empirical analysis. The revised manuscript first develops the theoretical logic of how different AI-answer signals should affect human entry and answer quality, and then tests these predictions empirically. This ordering makes clear that the empirical analyses are designed to evaluate theoretically specified mechanisms, rather than to rationalize results after the fact.

We hope the review panel finds that this revised theoretical narrative more clearly supports the paper's contribution.

References:

Chen, M.-J. (2009). Competitive dynamics research: An insider's odyssey. *Asia Pacific Journal of Management*, 26(1), 5–25.

Chen, M.-J., Su, K.-H., & Tsai, W. (2007). Competitive tension: The awareness–motivation–capability perspective. *Academy of Management Journal*, 50(1), 101–118.

Su, Y., Zhang, K., Wang, Q., & Qiu, L. (2025). Generative AI and human knowledge sharing: Evidence from a natural experiment [Working paper]. SSRN.

AE Comments

AE-0

> *As previously, the three reviewers once again did great jobs in objectively and constructively evaluating the revised paper. They commented on authors' responses to the previous suggestions and pointed out remaining and new issues. The three reviewers' comments are once again highly consistent. Unfortunately, instead of converging positively, reviewers remain concerned and critical in general. The collective recommendation from the reviewer panel is still negative, with only one reviewer indicating a further revision may be hopeful yet expressing more serious concerns than in the last round.*
>
> *I very carefully read the revised paper, the authors' revision notes, and the referee reports. I commend the authors' effort in a round of diligent revision, resulting in a much-improved paper. They addressed many of the issues highlighted in the SE and AE reports from last round. As a result, the concerns regarding the randomization have largely been eliminated; the main treatment effects have been shown stable when covariates are progressively added to the model; many data integrity questions have been clarified; the difference from existing literature has been discussed in detail.*

<u>Response</u>

We sincerely thank you for the thorough and careful evaluation of our revised manuscript, as well as the recognition of improvements in empirical analysis and literature review. We appreciate the review panel's further comments and we are committed to addressing these issues. We hope the review panel finds the paper significantly improved.

AE-1

> *Nonetheless, I have to agree with the reviewers that key concerns linger, namely, regarding mechanism analysis and theorization, which in essence go hand in hand.*
>
> *I believe the authors managed to establish that the main treatment effects are stable and robust, utilizing the unique opportunity of the randomized experiment of this focal platform. However, the review team feels that merely documenting such patterns, which in many ways contradict prior findings in the literature, is not enough to create meaningful new knowledge about what exactly is at play that leads to such phenomena. They need to be convinced of a plausible explanation of the underlying mechanism, supported by solid theories---ideally from a sound overarching theoretical framework. This paper appears to fall short in this regard unfortunately.*
>
> *Empirically, the analysis exploring the mechanism is somewhat flawed and hence unconvincing, as all three reviewers unanimously pointed out and I summarize in detail below. It then loops back to the issue of lacking a convincing theoretical framework, as all three reviewers complain. R2 questioned the approach here appears to be HARKing (i.e., hypothesizing after results are known). Such inadequacy further leads to the doubts on the discrepancy between the main effects found here and those reported in the literature (e.g., see R1's comments on Su et al, 2023).*

<u>Response</u>

We thank you for your comments. Following these comments and the SE's guidance, we have developed a new theoretical framework centered on the competitive encounter between human answerers and AI answers generated by the platform's internal GenAI system. Specifically, we theorize how the AI answer affects two decisions made by human answerers: the **entry decision**, or whether to contribute an answer at all, and the **crafting decision**, which concerns how good the human answer will be and how it will be produced, if the answerer decides to contribute.

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We link these signals to the two decisions through the two systems of thinking theory (Evans & Stanovich, 2013; Kahneman, 2003, 2011). The AI label and AI-answer length are **glance signals** processed through fast, low-effort System 1 thinking, whereas AI-answer quality is a **reading signal** processed through deliberate System 2 thinking. Similarly, entry is a System 1 decision because it can be settled quickly before close reading; the crafting decision is a System 2 decision because it is made after the answerer has committed to answering and determines both how good the submitted answer will be and how it will be produced.

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In line with this framework, the revised manuscript develops six hypotheses (see Figure AE-1 below): the AI label increases human answer supply; the effect of an AI answer on human answer supply becomes more negative as AI-answer length increases; AI-answer quality does not appreciably shape human answer supply; the AI label does not affect human answer quality; AI-answer length does not affect human answer quality; and the effect of an AI answer on human answer quality becomes more positive as AI-answer quality increases.

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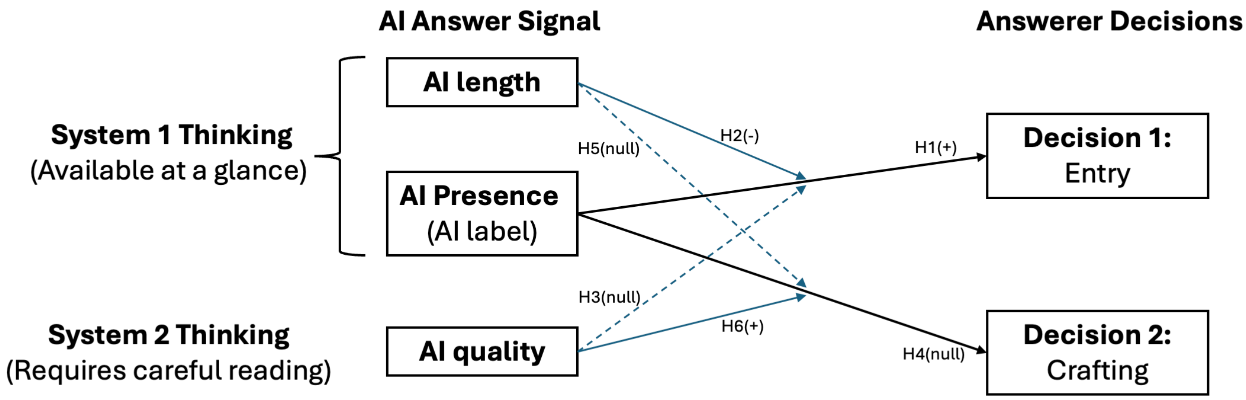

Figure AE-1. Conceptual framework

The revised framework also allows us to acknowledge alternative theoretical predictions and to reconcile our findings with prior work, including Su et al. (2025). Prior studies suggest that AI-generated answers can stimulate human participation rather than reduce it. Our framework accommodates this possibility by treating the AI answer not as a uniform treatment but as a profile of multiple signals with potentially offsetting effects. Consistent with this logic, we find that the AI label alone has a positive and significant effect on human answer supply (0.188, $p<0.05$), which is aligned with Su et al. (2025). However, the interaction with AI-answer length is negative and significant (-0.046, $p<0.001$). Thus, although the AI label may encourage entry, longer AI answers can deter entry by signaling that less room remains for a valuable human contribution. These countervailing effects help explain why the average total effect in our setting is negative.

We have also revised the manuscript to clarify why our setting differs from Su et al. (2025). We no longer frame the difference as one between technical experts and casual users. Instead, we emphasize differences in domain and deployment timing. Su et al. (2025) study a general-interest Q&A context, where answer value often rests on experience, interpretation, and personal perspective. In such settings, multiple answers can coexist, and an AI answer may function as an additional voice that stimulates further discussion. Our setting is a technical Q&A community, where answer value is more strongly tied to correctness, diagnostic precision, and actionable problem solving. In this context, a comprehensive AI answer is more likely to be viewed as occupying the solution space that a human answer might otherwise fill. In addition, in Su et al. (2025), AI answers were added to questions that already existed before the panel period and often already had human answers. In our field experiment, the AI answer appears within minutes of question posting and before any human answer arrives. Potential contributors therefore confront the AI answer at the moment of deciding whether to enter.

Together, these revisions address your concern that empirical analysis and theorization must go hand in hand. The revised manuscript first develops a coherent theoretical account of how internal GenAI changes human participation incentives and evaluation processes, and then tests predictions derived from that account. We hope you and the review panel find that the revised framework provides a more convincing explanation of the phenomenon, clarifies the relationship between theory and empirical analysis, and explains why our findings differ from, but can be reconciled with, prior work.

References:

Chen, M.-J. (2009). Competitive dynamics research: An insider's odyssey. *Asia Pacific Journal of Management*, 26(1), 5–25.

Chen, M.-J., Su, K.-H., & Tsai, W. (2007). Competitive tension: The awareness–motivation–capability perspective. *Academy of Management Journal*, 50(1), 101–118.

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AE-2

> *Mechanism driving reduction in human answers*
>
> *(a) Content effect*
>
> *As all three reviewers pointed out and I also agree, the fact that AI answers have higher quality than human answers has not been well established in the analysis of 6.3.1.1. The authors only show the direct testing result that AI answers have greater length than human answers, which is neither surprising nor supportive of the claim that AI answers have higher content quality than human answers.*
>
> *As R1 further observed, the focal platform has some guidelines against external GenAI tools, and users of this platform expressed substantial skepticism and dissatisfaction toward AI-generated answers in general. It further raises the question whether users indeed view AI-generated answers as trustworthy and having higher authority, which is used as the foundation of the whole mechanism analysis throughout the paper.*

Response

We thank you and the reviewers for prompting us to examine this issue more carefully.

Following this comment, we now conceptualize the AI answer as a profile of features, including the AI label, AI-answer length, and AI-answer quality. This revised approach allows us to distinguish the roles of AI-answer length and AI-answer quality in shaping human contributors' perceptions and behaviors.

Under the new framework, we theorize and show that human answerers in this community tend to perceive the AI as a relatively weak competitor, which draws them toward contribution rather than away from it (H1). At the same time, AI-answer length and AI-answer quality play distinct roles. A longer AI answer can deter human contribution by signaling that the answerable ground has already been substantially covered. By contrast, a high quality AI answer provides a foundation to build on, either by supplying adaptable material such as a correct diagnosis or working code that can be reused, or by charting a viable direction that the answerer can develop into a full answer. In this sense, AI content does not operate as a single mechanism. Instead, different content-related features can have different, and even opposing, effects.

By differentiating the AI label, AI-answer length, and AI-answer quality, the revised manuscript provides a more precise account of the AI content effect. It no longer relies on answer length as evidence of higher content quality. Instead, length is theorized as a coverage signal, while quality is measured and analyzed separately. We sincerely appreciate the review panel's suggestions, which helped us substantially improve the paper's theoretical clarity.

AE-3

> *(b) Position effect*
>
> *All three reviewers have issues with the approach surrounding "first-position seekers" for exploring the position effect. See R1's comments on page 4, R2's point 5 on the last page, and R3' comments in (b).*

Response

We thank you and the reviewers for the comments on the position effect. We agree that adequately testing this mechanism in our setting would require data on how answerers perceive and value answer position, which our current dataset does not provide. We have therefore removed the position effect from the manuscript in this revision. This change also helps us keep the revised framework more focused and ensures that the theoretical claims are directly supported by the available empirical evidence.

AE-4

> *(c) Source effect*
>
> *As all three reviewers questioned and I feel very much the same way, the analysis leading to the conclusion on the "source effect" of AI-generated answers appears to be the most unsound. Simply attributing all the remaining effects of AI answers on the decrease of human answers other than the (questionable) content and position effects to the "source effect" and interpreting it as a heightened perceived authority users give to AI answers seem too hasty and unjustified. On the very contrary, the large proportion of the unexplained residual effects (i.e., 54%) indicates there are very likely unmodeled alternative*

mechanism at work. See R1's last point on page 4, R2's comments on R2-20, and R3's comments in (c).

It becomes more questionable when considered in connection with the above points in (a) that AI answers may not necessarily have higher content quality and users may hold skepticism toward AI answers. The results in 7.1 that such authority effect does not exist for high-quality human answers further adds to the doubt about whether the perceived authority of AI answers really exists or not. As R2 rightly pointed out, it is unsubstantiated to extrapolate prior findings on users' trust on predictive AI to support the claim on users' trust on generative AI, because the latter may appear less reliable in nature due to issues like hallucination.

Response

Following the panel's suggestion, we have removed the residual-based attribution and the analogy to predictive AI. In the current version, we develop an overarching framework in which the effect of an AI answer depends on the AI answer profile. This framework separates the AI label from AI content and allows us to theorize their effects independently.

In this framework, we argue that seeing the AI label alone activates the community's prevailing domain-specific perception that AI answers could be unreliable for genuine technical problems. As a result, the label can lead human answerers to view the AI as a weak competitor and encourage contribution.

We hope this organization of the theory and evidence, under a single coherent framework, addresses the review panel's concerns.

AE-5

2. Mechanism driving reduction in viewer recognition

Reviewers are also unsold on the mechanism analysis leading to the reduction in viewer recognition. Both R2 (point 6 on the last page) and R3 (point d) questioned the attention diversion mechanism. R3 (point e) questioned the quality degradation mechanism. Both R1 (page 3) and R3 (point f) questioned the elevated expectation mechanism. As R1 pointed out, elevated expectation would make sense only if AI-generated answers are indeed of higher quality and regarded more trustworthy by users, which is an unsubstantiated assumption by itself and loops back to the discussion in (a) and (c) above.

Response

We thank you and the reviewers for raising concerns about the mechanisms underlying the reduction in viewer recognition. We carefully investigated these concerns and conducted the additional analyses requested by the reviewers. Although we can provide evidence that addresses several of these concerns, we also recognize that some uncertainty remains.

In light of these issues, and consistent with the SE's suggested framing, we have removed the analyses of viewer recognition from the manuscript. At the same time, we retain the findings that are directly relevant to the revised framework, which focuses on human answerers' entry and crafting decisions. We appreciate the reviewers' suggestions, which helped us sharpen the theoretical scope of the paper.

AE-6

> *In addition, reviewers also pointed out some remaining data integrity issues, for example, R1's first comment on page 4, R2's comments on R2-7, R2-8, and R-16.*

Response

We thank you for consolidating these points. We reviewed each one carefully and confirm that they concern reporting and typographical issues rather than the underlying analyses. We have addressed all of them. Specifically, we corrected the typos noted in R1-11 and R1-18, conducted a comprehensive proofreading of the entire manuscript to remove similar inconsistencies, improved the transparency of our reporting in response to R2-7 and R2-8, and corrected the issue raised in R2-16.

The detailed corrections are provided in our responses to R1-11, R1-18, R2-7, R2-8, and R2-16.-

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AE-7

> *As much as I hate to see papers get rejected in the second round, I also hate to have the review process dragging on without a clear direction to pursue and a promising end in sight. Given that pretty much the entire review team remain concerned and negative about issues as fundamental as discussed above in the second round, after reflecting on reviewers' comments and deliberating for a long while, I feel it is probably in the best interest of both the authors and the review team to call it quits at this point.*
>
> *This being said, I would certainly encourage the authors not to give up on this research, especially the valuable data from this unique randomized field experiment. They just need to spend more time to think about and explore deeply what exactly are the mechanisms at work leading to the patterns observed on the surface and how such mechanisms would reconcile with existing theories and prior findings. Only so could the interesting observations in the data be better understood/trusted and converted into meaningful new knowledge. I wish the authors the best!*

Response

We sincerely thank you for the thoughtful assessment of our manuscript and for the constructive guidance on how it might be strengthened. We appreciate the recognition that the randomized field experiment provides valuable data and may support a meaningful contribution. We also take seriously the concern that the manuscript needs a clearer theoretical account of the mechanisms underlying the observed effects and of how those effects relate to existing theory. In the revision, we have focused on developing that account more fully and have designed the empirical analyses to target the proposed theoretical argument more directly.

Reviewer 1 Comments

R1-0

> *Thank you for revising and resubmitting the paper. Although I appreciate the authors' efforts to address my earlier comments, I find that the manuscript still has several major issues that limit its theoretical and practical contribution. I outline below a set of comments that I hope will help the authors further improve the rigor, clarity, and overall contribution of the work.*

Response

We sincerely thank you for the continued engagement with our manuscript and the recognition of our revision efforts. We appreciate your commitment to helping us improve the rigor, clarity, and overall contribution of this work. The detailed comments provided offer valuable guidance for addressing the remaining issues.

R1-1

> *– In the opening paragraphs of the introduction, the authors distinguish internal from external GenAI largely from a researcher's perspective (e.g., data observability, attribution challenges, experimental feasibility). While these points explain why the setting is methodologically attractive, they do not sufficiently motivate why the phenomenon itself is theoretically important. What ultimately matters is not that internal GenAI is easier for researchers to study, but that it represents a fundamentally different sociotechnical and governance arrangement from external GenAI. The former involves platform control, explicit labeling, institutional endorsement, and altered legitimacy and accountability. I encourage the authors to reframe the motivation around these substantive differences and use the data-access advantages as supporting justification rather than the primary motivation for the research question.*

Response

We thank you for this insightful comment, and we fully agree. In the revision, we have reframed the motivation around the substantive sociotechnical and governance differences between internal and external GenAI, and demoted the data-access advantages to a supporting justification, exactly as suggested.

In the revised introduction, we now argue that internal GenAI is not simply external GenAI relocated inside the platform but a distinct sociotechnical and governance arrangement. We make three substantive differences explicit. First, on platform control, the platform rather than an outside vendor decides whether an AI answer appears, how it is presented, and how capable it is, so its presence becomes a design choice rather than an environmental shift. Second, on explicit labeling and institutional endorsement, the label carries an institutional signal about who produced the answer and reshapes how the community assesses its legitimacy and reliability. Third, on altered legitimacy and accountability, because the AI competes for the same readers as any human answer, it enters the community as an identified competitor rather than a private substitute. This reframing also grounds our theoretical framing: whereas external GenAI has been treated as a private substitute, the internal answerer enters the community as a visible answerer competitor, which is the foundation of the competition-based framework we develop in this revision.

Consistent with your guidance, we now introduce the data-access advantages only after this substantive motivation and present them as supporting justification. Because the platform labels the AI answer and randomly assigns which questions receive one, our setting lets us observe and isolate an AI effect that remains hidden when GenAI is consulted privately.

Please see the revised Introduction for these changes.

R1-2

> *– I appreciate the authors for providing additional contextual details, such as the platform's announcement about the restricted use of external GenAI tools (https://segmentfault.com/a/1190000043031126). I recommend the authors including these details in the paper due to their critical importance.*

Response

Thank you for the comment. We have now incorporated the announcement link into Appendix B where detailed background on the randomized experiment is provided.

R1-3

> *– In the response letter, the authors note that SegmentFault instituted policies discouraging or restricting the use of external GenAI tools. Yet, the revised manuscript repeatedly frames the contribution as estimating the incremental effect of internal GenAI in an environment where users have access to external tools like ChatGPT. These two claims appear in tension. If external GenAI use is meaningfully restricted, then the baseline environment is not one of "external GenAI availability," and the interpretation and generalizability of the findings change. The authors should clarify the extent and enforcement of this policy and align the paper's framing and claimed contribution accordingly.*

Response

We thank you for raising this, and we appreciate the opportunity to resolve the apparent tension you identify. The tension dissolves once the scope of the community's policy is made precise, and we agree that the manuscript needed to state this scope clearly and align its framing accordingly.

The community's restriction targets posting AI-generated answers, that is, submitting output copied from an external tool as one's own work on the platform. It does not target, and in practice cannot detect or prevent, a user privately consulting ChatGPT or learning from it before deciding whether and how to answer. Therefore, the policy governs what appears in the answer thread, not what users do before they post. This is an important distinction for your concern: the baseline environment is still one of external GenAI availability, because the demand-diverting channel that prior work documents, that is, users obtaining answers privately from external tools rather than turning to the community, operates entirely upstream of any posting and is untouched by the restriction. What the policy removes is only the contamination of our outcome measures by copied external output.

We have made this explicit in two places. In Appendix B (Background on the Randomized Experiment and Randomization Check), we now state: “SegmentFault implemented a policy prohibiting the posting of AI-generated answers, that is, answers that a user copies from an external tool and submits as the user’s own work. It does not, and in practice cannot, prevent users from consulting ChatGPT or learning from it before they post” We then define the baseline precisely in the Introduction: “Because external GenAI remained available for private use in both conditions and continued to divert demand from the community, our estimates capture the incremental effect of adding internal GenAI rather than the effect of GenAI relative to an AI-free environment.”

R1-4

> *– A related concern is the apparent inconsistency between the platform’s restriction of external GenAI and its later deployment of its own internal GenAI. According to the policy link cited in the response letter, the restriction on tools such as ChatGPT is motivated largely by concerns about inaccuracy and hallucination. This raises an important question: why would users view the platform’s internal GenAI as sufficiently reliable to warrant prominent first-position answers, when external GenAI is discouraged for reliability reasons? From a user perspective, this asymmetry may undermine trust or generate frustration, potentially discouraging participation for reasons unrelated to the content, source, or position mechanisms proposed in the paper. The authors should clarify how the platform justified this distinction to users and discuss whether governance inconsistency or user resentment could contribute to the observed effects, as well as what this implies for the generalizability of the findings.*

Response

We thank you for this thoughtful comment. We understand the concern as raising two related questions: first, whether the platform’s restriction on external GenAI and its deployment of an internal GenAI system create a governance inconsistency; and second, whether users may have reacted negatively to this perceived inconsistency in ways that could confound the mechanisms originally proposed in the paper.

To clarify, the two policies are actually coherent. The restriction on external GenAI use and the deployment of internal GenAI are responses to the same governance problem: how to manage AI-generated content in a technical Q&A community where accuracy, accountability, and source transparency are central. Unrestricted use of external GenAI would allow many users to post AI-generated answers as if they were their own, without consistent disclosure of the source. This would make it difficult for readers to identify which answers were AI-generated and to calibrate their trust accordingly. By contrast, the platform’s internal GenAI answer is displayed with a clear AI label. From a governance perspective, the distinction is therefore not simply “external AI prohibited, internal AI allowed.” Rather, it is a distinction between many undisclosed AI-generated answers flooding the platform and a single, visibly labeled AI answer whose source is transparent to users.

If users responded with strong frustration or resentment toward the internal GenAI deployment, we would expect the effect to appear in broader engagement outcomes, not only in the decision to contribute an answer after seeing a specific AI response. For example, users might stop posting questions, visit fewer question pages, reduce browsing activity, or engage less in follow-up dialogue within threads. However, the analyses reported in the previous version of the manuscript show no such broad decline: AI-generated answers have neutral effects on questioner retention, interactive dialogue, and question viewership. These patterns suggest that the intervention did not meaningfully reduce general platform engagement.

In addition, we conducted a systematic search of user posts on the platform for expressions of frustration, resentment, or opposition toward the internal GenAI system. We did not find evidence of such user backlash. Therefore, a broad resentment-based explanation is less consistent with the available data.

R1-5

> *– I appreciate the authors' efforts in differentiating from Su et al. (2023). However, I do not find the listed differences convincing. The authors should reconsider or substantially revise their explanation for why their findings diverge from Su et al. (2023).*
>
> *The claim that Quora is a "general-interest" platform whereas the present study focuses on technical users is overstated. Quora does host IT- and technology-focused communities, and Su et al. explicitly analyze objective versus subjective questions and expert versus non-expert contributors. Thus, differences in user composition alone cannot explain the opposite findings.*
>
> *In the response letter and revised paper, the authors repeatedly claim that in Su et al. (2023), AI-generated answers arrive "approximately three months after a question is posted." I could not find support for this statement. In Su et al., the three-month period refers to the observation window before and after the policy launch, not to a delay between question posting and AI answer arrival.+*
>
> *More importantly, even if one were to assume a delay, it is unclear why this would fundamentally alter contributor incentives. Su et al. conduct a controlled online experiment that holds timing constant and still finds positive effects of AI answers. Moreover, human answerers rarely respond immediately after question posting; what matters is whether an AI answer is visible and salient at the time contributors consider answering. As long as the AI answer occupies a prominent position in the answer section, it should affect human contribution decisions regardless of whether it appears immediately or later.*

Response

We agree that user composition alone cannot explain the opposite findings, and we have shifted our explanation to the contribution environment. We no longer claim that Quora lacks technical or objective questions. The more relevant difference is the type of contribution each platform elicits. Prior work suggests that Quora carries stronger conversational elements than purely technical Q&A communities such as Stack Overflow (Oliveira et al., 2018). In a conversation-oriented environment, an AI answer can serve as an additional reference point in an ongoing discussion, so contributors may perceive it as complementary to human participation. By contrast, in a technical Q&A community such as SegmentFault, a question typically calls for a correct diagnosis or an executable solution, so an AI answer is more likely to occupy the same solution space that a human answer would otherwise fill. In that setting, the AI answer is more likely to be perceived as a competitor to human answerers. We have revised the Research Gap and Conclusion sections to make this distinction clearer.

We also thank you for correcting our interpretation of the timing in Su et al. (2025). The three-month period refers to the observation window before and after the policy launch, not to a fixed delay between question posting and AI answer arrival. After reviewing the most recent version of Su et al. (2025), we found that timing remains an important distinction between the two studies. In Su et al. (2025), the AI answer is introduced long after the original question was posted (average pre-policy tenure is roughly 49 months), by which point at least one human answer is already present (Su et al. 2025, pp. 15, 22). The AI therefore enters as a retrospective supplement to an established discussion. In our setting, the AI answer is generated quickly after the question posting and before human contributors arrive, so it is the rival that a contributor evaluates at the moment of entry.

Regarding your comment that "*As long as the AI answer occupies a prominent position in the answer section, it should affect human contribution decisions regardless of whether it appears immediately or later,*" our point is that the nature of the AI answer effect and the estimate one obtains from the sample critically depend on the timing of the AI answer arrival. Consider a mechanistic situation where an older and crowded question was already receiving few new human answers, so the arrival of an AI answer has little inflow to impact even if the AI answer occupies a prominent position, whereas a fresh question still attracting answers gives the AI answer plenty of room to impact.

Our argument is that the timing of the AI answer arrival impacts how human answerers view the AI answer. When the AI answer arrives first into an empty question, it defines the answer and occupies the role a human would have played. Human answerers are therefore more likely to perceive it as a competitor. When it arrives after humans have already staked out the thread, it looks like an add-on to a human-led discussion. Human answerers are more likely to perceive it as a complement.

We note that Su et al. (2025) conducted a lab experiment, but it examines a different setting, and its design does not address exactly the same research question as ours. **First**, both conditions in their experiment already contain three human answers alongside the AI answer, so the AI is a supplement layered on an existing human-answer stock rather than a first mover on an empty question. **Second**, every participant is required to write an answer, which removes the decision of whether to contribute at all. Su can therefore observe only how long already-committed respondents write and how closely they track the AI, not whether the AI deters potential contributors from answering. **Third**, the experimental questions are drawn from a Quora dataset, and the only disclosed example is a parenting question ("What's the best parenting advice?"). Technical questions would require respondents with relevant expertise, and no such expert recruitment is reported, so the type of question and the type of contributor both differ from our setting.

Finally, our revised theoretical framework helps reconcile our findings with those of Su et al. (2025). We now theorize that an AI answer consists of multiple observable features, including the AI label, AI-answer length, and AI-answer quality. These features can have different effects. In our results, the effect of the AI label alone on human answer supply is positive and significant (0.188, p<0.05), which is consistent with Su et al.'s finding that AI answers can stimulate participation. However, the interaction between the AI label and AI-answer length is negative and significant (-0.046, p<0.001). This pattern suggests that while the AI label may invite contribution, longer AI answers can deter contribution because they signal that the question has already been substantially covered. The average effect becomes negative in our setting because the deterrent effect of apparent coverage offsets and ultimately dominates the positive label effect.

We hope this revised framing provides a clearer and more convincing explanation for the difference between the two sets of findings.

References:

Oliveira, N., Muller, M., Andrade, N., & Reinecke, K. (2018). The exchange in StackExchange: Divergences between Stack Overflow and its culturally diverse participants. Proceedings of the ACM on Human-Computer Interaction, 2(CSCW), Article 159, 1–22.

Su, Y., Zhang, K., Wang, Q., & Qiu, L. (2025). Generative AI and human knowledge sharing: Evidence from a natural experiment [Working paper]. SSRN.

R1-6

> *– I have concerns about the theoretical coherence of Section 3.1, particularly regarding the source effect and the use of dual-process theory.*
>
> - *First, the source effect relies on the assumption that users attribute heightened authority and credibility to AI-generated answers, drawing on algorithm appreciation and automation bias. However, this assumption appears to conflict with the institutional context. The platform explicitly restricted external GenAI use due to concerns about inaccuracy. After carefully checking the translated version of user comments under the announcement (https://segmentfault.com/a/1190000043031126), it appears that users expressed substantial skepticism and dissatisfaction rather than unquestioned trust toward AI-generated answers.*

<u>Response</u>

We thank you for raising this issue, which prompted us to reconsider our theorization. We examined the tension you identified closely and also reviewed the literature on content generated by GenAI (Kabir et al. 2024, Longoni et al. 2022). Consistent with your observation, this literature suggests a general skepticism toward GenAI-generated content, particularly in technical domains.

Our revised theoretical framework decomposes the AI effect into a profile of AI features: label, length, and quality. We argue that human answerers view GenAI as a weak competitor, so knowing that an answer is authored by GenAI encourages human contribution. Our results support this account: the effect of the AI label alone on supply is positive and significant (0.188, $p < 0.05$). At the same time, the interaction between the AI indicator and AI-answer length is negative (-0.046, $p < 0.001$), indicating that a longer AI answer deters human answerers. This length effect drives the total effect to be negative, as we identified in previous versions of the paper.

In summary, we agree that the platform context does not support an assumption of unquestioned trust in AI-generated answers. In this technical Q&A community, the AI label may trigger skepticism rather than trust. We elaborate on this reasoning for H1 in Section 4.

References:

Kabir, S., Udo-Imeh, D. N., Kou, B., & Zhang, T. (2024). Is Stack Overflow Obsolete? An Empirical Study of the Characteristics of ChatGPT Answers to Stack Overflow Questions. In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 1–17.

Longoni, C., Fradkin, A., Cian, L., & Pennycook, G. (2022). News from Generative Artificial Intelligence Is Believed Less. In Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency, 97–106.

R1-7

- *Second, the attempt to integrate the content, source, and position effects using dual-process theories (e.g., central vs. peripheral routes) feels underdeveloped and somewhat stretched. It is unclear why specific effects should operate through particular cognitive routes, and no empirical evidence or established literature is provided to justify these mappings. Dual-process theories typically require clear conditions under which central versus peripheral processing dominates (e.g., motivation, ability, involvement), which are not articulated or tested here. As currently presented, the framework appears post hoc. The authors should either provide stronger theoretical and empirical justification for this integration or simplify the theorization to avoid overextension.*

Response

Following your suggestion, we have revised the theoretical development section, with one central goal being to simplify the model and tighten its focus. Rather than mapping the content, source, and position effects onto separate cognitive routes, the revised framework organizes the theorization around a leaner, two-system account of the answerer's decision.

Specifically, we theorize how the AI answer affects two decisions made by human answerers: the **entry decision**, or whether to contribute an answer at all, and the **crafting decision**, which concerns how good the human answer will be and how it will be produced, if the answerer decides to contribute.

We adopt a competition lens informed by competitive dynamics theory (Chen et al., 2007; Chen, 2009). This lens matches the setting because the AI answer sits in the same thread and on the same screen as any human answer that follows. A later human answer must therefore be worth reading in the presence of the AI answer. It also gives us richer predictions than a simple substitution account, which would predict only that AI answers reduce human contributions. By contrast, the competition lens explains why an AI answer may discourage human entry in some cases while eliciting higher-quality human answers from those who do enter.

The revised framework treats the AI answer as a profile of three signals:

(a) **AI label.** The label identifies the source of the existing answer and tells the answerer what kind of competitor this is. Because machine authorship is apparent at a glance, the label can shape the answerer's initial appraisal before the AI answer's content is read. In our theory, the label leads answerers in this technical Q&A setting to appraise the AI as a weak competitor, which strengthens the incentive to enter.

(b) **AI-answer length.** Length is also available at a glance. It signals how much of the answerable ground the AI answer appears to have taken. A longer AI answer can appear to cover more possible causes or fixes, which reduces the perceived room left for a human contribution and weakens the incentive to enter.

(c) **AI-answer quality.** Quality is not available at a glance. It becomes available only when the answerer reads the AI answer and evaluates whether its reasoning, diagnosis, or code fits the specific problem. Once a human answerer has decided to contribute, a higher-quality AI answer serves as a better foundation to build on, either by supplying adaptable material such as a correct diagnosis or working code that can be reused, or by charting a viable direction that the answerer can develop into a full answer.

We link these signals to the two decisions through the two systems of thinking theory (Evans & Stanovich, 2013; Kahneman, 2003, 2011). The AI label and AI-answer length are **glance signals** processed through fast, low-effort System 1 thinking, whereas AI-answer quality is a **reading signal** processed through deliberate System 2 thinking. Similarly, entry is a System 1 decision because it can be settled quickly before close reading, the crafting decision is a System 2 decision because it is made after the answerer has committed to answering and determines both how good the submitted answer will be and how it will be produced.

This classification yields our **same-level principle**: a signal affects a decision when both occupy the same level of processing, while cross-level effects are expected to be null. Thus, the AI label and AI-answer length shape entry, but AI-answer quality does not. Conversely, AI-answer quality shapes human answer quality by serving as a foundation the answerer can reuse and redirect to lower the cost of producing a high-quality answer, but the AI label and AI-answer length do not systematically shape human answer quality once the answerer has entered.

In line with this framework, the revised manuscript develops six hypotheses (see Figure R1-7 below): the AI label increases human answer supply; the effect of an AI answer on human answer supply becomes more negative as AI-answer length increases; AI-answer quality does not appreciably shape human answer supply; the AI label does not affect human answer quality; AI-answer length does not affect human answer quality; and the effect of an AI answer on human answer quality becomes more positive as AI-answer quality increases.

We then test these predictions using our empirical data. In light of this revised conceptual framework, we have removed the analysis of viewer recognition and the position effect to keep the paper theoretically focused and internally coherent.

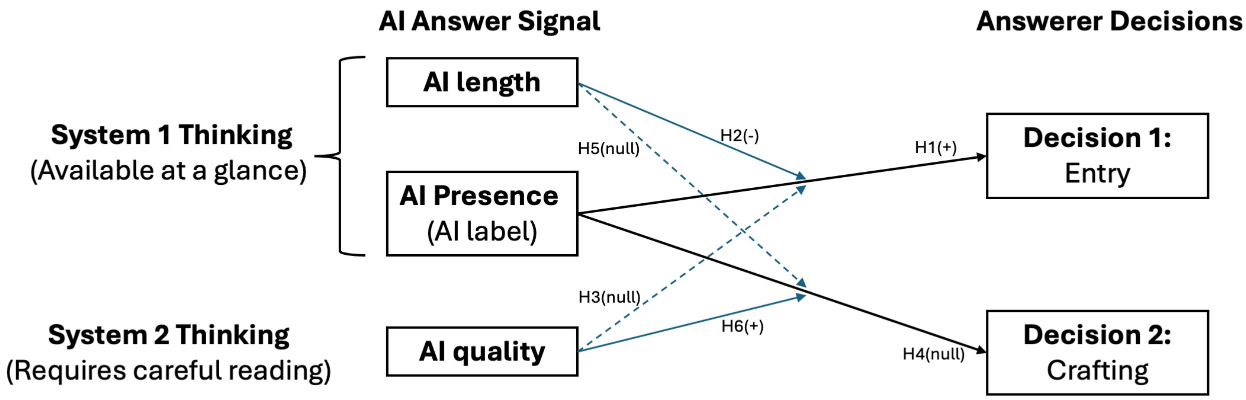

Figure R1-7. Conceptual framework

By tying each feature to the two systems of thinking, this streamlined framework makes the mapping between cognitive routes and effects explicit. We hope the revised theoretical development section is both simpler and cleaner in its theoretical grounding.

References:

Chen, M.-J. (2009). Competitive dynamics research: An insider's odyssey. *Asia Pacific Journal of Management*, 26(1), 5–25.

Chen, M.-J., Su, K.-H., & Tsai, W. (2007). Competitive tension: The awareness–motivation–capability perspective. *Academy of Management Journal*, 50(1), 101–118.

Evans, J. St. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, 8(3), 223–241.

Kahneman, D. (2003). A perspective on judgment and choice: Mapping bounded rationality. *American Psychologist*, 58(9), 697–720.

Kahneman, D. (2011). Thinking, fast and slow. Farrar, Straus and Giroux.

R1-8

– *I find Section 3.2 conceptually problematic.*

- *The distinction between the proposed mechanisms, expertise composition shift and reduced effort investment, is unclear. The expertise composition shift assumes that AI answers disproportionately discourage experts while having minimal impact on non-experts, but the paper does not explain why non-experts would not also be discouraged by the same signals of redundancy or elevated standards. The reduced effort investment mechanism further blurs this distinction, as both mechanisms ultimately describe contributor discouragement leading to lower-quality human answers. As currently theorized, these mechanisms are not analytically separable.*
- *The elevated expectations mechanism again relies on the assumption that users perceive AI-generated answers as superior and norm-setting. This assumption is not well*

supported in the platform context, where AI use is explicitly restricted due to accuracy concerns and user reactions suggest skepticism toward AI-generated answers. Without clearer justification that AI establishes a credible quality benchmark, the elevated expectations mechanism remains speculative.

Response

In light of your concern and the SE's suggestion regarding the framing of the paper, we have removed the analyses on view recognition. This change makes the paper more focused and ensures that our empirical analyses are more specifically targeted at the theoretical argument we intended to test. Because the mechanisms you raise (expertise composition shift, reduced effort investment, and elevated expectations) all underlie view recognition, they are no longer theorized or tested in the revised paper. We thank you for these suggestions, which have helped us improve the paper.

R1-10

- *In Section 4.1, the authors assert that AI-generated answers "typically occupy the first position for treated questions" and that viewers "typically see this content before human-generated answers." It is unclear what "typically" means in practice. Does this imply that human-generated answers sometimes occupy the first position despite the presence of AI-generated answers? If so, the authors should report descriptive statistics on answer ordering, such as the proportion of treated questions in which the AI answer appears before any human answer and the fraction of cases in which human answers precede the AI answer.*

Response

We apologize for the confusion. To clarify: on this platform, an AI-generated answer is, by design, always positioned before any human-generated answer for a treated question. The AI answer is not interleaved with human answers according to posting time; instead, it is anchored to the top of the answer list. We have further elaborated this in Section 4.1.

R1-11

– I noticed inconsistencies in the reporting of key covariates across the main tables and the appendix. For example, Table 2 reports the average question length for the treatment group as 84.233 Chinese characters, whereas Table A1 reports a mean "log number of Chinese characters" of 0.004; similarly, Table 2 reports an average questioner reputation of 334.484, while Table A1 reports a mean "log reputation score" of 4.026. These values are not consistent under standard log transformations.

Response

We apologize for this confusion. Upon careful review, we identified a formatting error in the Appendix where values were misaligned across columns. We have thoroughly corrected all such inconsistencies. Thank you for your careful reading that helped us identify this issue.

R1-12

> *– Section 6.3.1.1 uses "first answer length" as a key variable in testing the content effect, but the definition is unclear. In treated questions, does this refer to the length of the AI-generated answer (which is typically the first answer), or to the length of the first human answer? These two interpretations have very different implications for the mechanism test and for causal interpretation, especially given that human answer length would be post-treatment. The authors should clearly define this variable and justify its use in the analysis.*

Response

We apologize for the confusion. "First answer length" refers to the length of the first answer to a question, which could be either an AI answer or a human answer: in the treatment group, it is the length of the AI answer, whereas in the control group, it is the length of the human answer.

R1-13

> *– I find the logic surrounding "first-position seekers" in Section 6.3.1.1 unconvincing. The authors argue that when an AI answer occupies the first position, first-position seekers are less likely to provide the first human answer. However, even in the presence of an AI answer, users can still be the first human responder, so it is unclear why this opportunity is meaningfully reduced. This interpretation implicitly assumes that first-position seekers care only about being absolutely first rather than first among human contributors, which is neither theoretically justified nor empirically demonstrated.*
>
> *Moreover, the finding that the presence of a first-position seeker strongly stimulates subsequent participation is counterintuitive. Early answers often satisfy demand and crowd out later contributions rather than encouraging them. The paper does not explain why first-position seekers would systematically generate answers that invite further engagement. Are answers from these first-position seekers of low quality?*

Response

We appreciate these concerns and agree with them. As you note, positing that first-position seekers care only about being absolutely first, rather than about being first among human contributors, is a strong assumption. Even when an AI answer occupies the first position, users can still be the first human responder, so the opportunity that first-position seekers value is not meaningfully reduced. We also agree that the claim that first-position seekers stimulate subsequent participation is difficult to reconcile with the pattern in which early answers satisfy demand and crowd out later contributions. Given the weak theoretical footing and the counterintuitive implications you and other reviewers highlight, we have removed the first-position-seeker argument from our theorization and revised the paper accordingly.

R1-14

> *– In Section 6.3.1.3, the authors state that they control for the position effect when estimating the source effect, but it is unclear what specific variables constitute the "position effect controls" and how they capture answer ordering or visibility.*
>
> *More importantly, the claim that a remaining significant coefficient on the AI indicator*

constitutes evidence of a source effect is too strong. A residual AI effect could reflect other unmodeled mechanisms correlated with AI presence, such as user discouragement or frustration induced by the platform's restrictive policy toward external GenAI and negative user reactions to AI deployment. Without ruling out such alternative explanations or independently manipulating source cues, the residual treatment effect cannot be uniquely attributed to a source effect.

<u>Response</u>

We thank you for this comment. In the earlier version, we tested the position effect through a proxy based on user behavior, that is, whether the first human answer came from a "first-position seeker."

In this revision, in light of your concern about the position effect and the SE's framing suggestion, we have removed the position effect from the framework. As a result, the "position effect controls" no longer appear in the revised paper, and the framework is now organized around effects we can examine directly.

R1-15

– I recommend reporting the results of the causal mediation analysis in a table and provide enough details about the implementation.

<u>Response</u>

In the previous version of the manuscript, we implemented an Imai-Keele-Tingley-style causal mediation analysis (Imai et al., 2010). In the revised manuscript, we no longer use causal mediation analysis to investigate the mechanisms, for two reasons. First, causal mediation analysis requires us to measure the theorized mediators, but our data do not allow us to observe or measure them. Second, the Imai-Keele-Tingley approach relies on a strong assumption (sequential ignorability), which requires that both x and m in the causal chain x->m->y be randomized. This assumption is unlikely to hold in our setting, because the mediators were not randomized.

The mechanism analysis in the revised manuscript instead explores the mechanisms that may underlie our hypotheses. The goal is not to estimate the magnitude of each mechanism, as causal mediation analysis would, but to test their observable implications. Each mechanism implies patterns that should appear in the data if it is operating, and these patterns often differ from those implied by competing explanations. We test for these patterns and use them, where possible, to distinguish our account from alternatives. These results should therefore be read as support for the plausibility of our arguments, not as estimates of the magnitude of each channel.

Reference:

Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, 15(4), 309–334.

R1-16

– In Section 6.3.2.1, the authors use comment volume to rule out attention diversion. While this is informative, under the elevated expectations mechanism, comments may become more critical or negative rather than fewer. Comment sentiment, not just volume, would therefore be more informative.

Response

Thank you for this thoughtful suggestion. We have conducted the analysis using comment sentiment, and we did not find that viewers express significantly more critical or negative sentiment toward human answers in the presence of AI.

As shown in Table R1-16 below, the coefficients on AI are insignificant under both measures, indicating that viewers do not express more critical sentiment toward human answers when an AI answer is present.

Table R1-16. Effect of AI Answers on the Number of Negative Comments Received by Human Answers

	(1)	(2)
DV:	$negCommentNum\_cnsenti_j$	$negCommentNum\_snowNLP_j$
AI_i	0.061	0.001
	(0.037)	(0.084)
other controls	√	√
answer timing FE	√	√
N	1,451	1,451

That said, in light of your concern about our original theoretical framing and the SE's framing suggestion, we have removed the analyses on viewer recognition from the paper. We report this analysis here for transparency, and we thank you for the suggestion.

R1-17

– A broader concern is that several analyses throughout the paper restrict attention to subsamples of questions with at least one human answer. Because the main treatment effect is a reduction in human answers, conditioning on having an answer constitutes post-treatment selection. This restriction disproportionately excludes treated questions and selects on the outcome variable, potentially biasing estimates and complicating interpretation of mechanism tests. The authors should clearly justify these sample restrictions, discuss the resulting selection, and assess whether the conclusions drawn from these subsamples generalize to the full set of treated questions.

Response

We thank you for raising the issue of post-treatment selection. In the revised manuscript, our main analysis of the effect of AI answers on human answer quality also conditions on the presence of a human answer. To address the potential bias arising from this post-treatment selection, we estimate Lee bounds using two different approaches (Lee 2009). The resulting bounds are tight, contain our point estimates, and exclude zero (See Appendix D for details).

Post-treatment selection is therefore unlikely to change our conclusions.

Reference:

Lee, D. S. (2009). Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects. *The Review of Economic Studies*, 76(3), 1071–1102.

R1-18

– Page 19 of the revised manuscript, there is a typo: "Summary StatisticsTable 2 ".

Response

Thank you for catching this typo. We have corrected it and conducted a comprehensive proofreading of the entire manuscript to ensure no similar issues remain.

R1-19

I hope the authors find these comments helpful. Best of luck!

Response

We are deeply grateful for your thorough and constructive review. Your comments have been invaluable in strengthening both the rigor and clarity of our manuscript. We sincerely appreciate the time and effort you have devoted to improving this work.

Reviewer 2 Comments

R2-0

Thank the editors for inviting me to review the revised version of this paper. Thank the authors for addressing my earlier comments.I am the R2 in the last round of review. My process for this round review is as follows. First, I scan the authors' responses. Second, I re-read the revised paper carefully. Third, I carefully re-read each of my earlier comments and each response from the authors, and consider whether my comments are addressed. If they are addressed, I label them as addressed. If not, I will provide further elaboration/comments. To organize the work, I followed the same comment label from the authors.

Since this is a revised paper, and I read the paper to understand how my earlier comments were addressed, I, therefore, have some new comments. At the same time, this is a revision. Hence, I keep new comments to a minimum. I hope my review process makes sense to the authors and editors. I hope the authors the best of luck in improving their paper.

Response

We sincerely thank you for the systematic and constructive comments. We have carefully revised the paper to address your comments. We hope you find the paper satisfactory this time.

R2-1

Theoretical Foundation

R2-1. In the earlier round, I asked why the authors focused on content effect, source effect, and position effect instead of other theoretical mechanisms such as the learning effect and the crowded-out effect.

The authors responded by showing that the literature reveals that the three effects are used to explain how answers (earlier answers) affect subsequent user participants, attention, and perceived credibility. And the authors integrate their knowledge of the research context with those established theories/literature findings.

It may not be a good way to justify the selection of the theoretical mechanisms. Other mechanisms, such as the learning, crowd-out, anchoring effect, and cognitive offloading (can be more), are also examined in the literature and can be used to explain this research context (i.e., AI answers' effect on human answers).

Probably, the authors can find an overarching theoretical framework to support their exploration of what the AI says (content effect), who says it (whether AI or human; source effect), and when/where the AI says it (position effect).

Response

We thank you for your suggestions on the choice of mechanisms. Following the overarching framework suggested by the SE, we now model the AI answer as a competitor to human answers. The revised theoretical model incorporates many of the insights you have offered across the two rounds of review.

We have developed a new theoretical model centered on the competitive encounter between human answerers and AI answers generated by the platform's internal GenAI system. Specifically, we theorize how the AI answer affects two decisions made by human answerers: the **entry decision**, or whether to contribute an answer at all, and the **crafting decision**, which concerns how good the human answer will be and how it will be produced, if the answerer decides to contribute.

We adopt a competition lens informed by competitive dynamics theory (Chen et al., 2007; Chen, 2009). This lens matches the setting because the AI answer sits in the same thread and on the same screen as any human answer that follows. A later human answer must therefore be worth reading in the presence of the AI answer. It also gives us richer predictions than a simple substitution account, which would predict only that AI answers reduce human contributions. By contrast, the competition lens explains why an AI answer may discourage human entry in some cases while eliciting higher-quality human answers from those who do enter.

The revised framework treats the AI answer as a profile of three signals:

(a) **AI label.** The label identifies the source of the existing answer and tells the answerer what kind of competitor this is. Because machine authorship is apparent at a glance, the label can shape the answerer's initial appraisal before the AI answer's content is read. In our theory, the label leads answerers in this technical Q&A setting to appraise the AI as a weak competitor, which strengthens the incentive to enter.
(b) **AI-answer length.** Length is also available at a glance. It signals how much of the answerable ground the AI answer appears to have taken. A longer AI answer can appear to cover more possible causes or fixes, which reduces the perceived room left for a human contribution and weakens the incentive to enter.
(c) **AI-answer quality.** Quality is not available at a glance. It becomes available only when the answerer reads the AI answer and evaluates whether its reasoning, diagnosis, or code fits the specific problem. Once a human answerer has decided to contribute, a higher-quality AI answer serves as a better foundation to build on, either by supplying adaptable material such as a correct diagnosis or working code that can be reused, or by charting a viable direction that the answerer can develop into a full answer.

We link these signals to the two decisions through the two systems of thinking theory (Evans & Stanovich, 2013; Kahneman, 2003, 2011). The AI label and AI-answer length are **glance signals** processed through fast, low-effort System 1 thinking, whereas AI-answer quality is a **reading signal** processed through deliberate System 2 thinking. Similarly, entry is a System 1 decision because it can be settled quickly before close reading, the crafting decision is a System 2 decision because it is made after the answerer has committed to answering and determines how good the submitted answer will be and how it will be produced.

This classification yields a **same-level principle**: a signal affects a decision when both occupy the same level of processing, while cross-level effects are expected to be null. Thus, the AI label and AI-answer length shape entry, but AI-answer quality does not. Conversely, AI-answer quality shapes human answer quality by serving as a foundation the answerer can reuse and redirect to lower the cost of producing a high-quality answer, but the AI label and AI-answer length do not systematically shape human answer quality once the answerer has entered.

In line with this framework, the revised manuscript develops six hypotheses (see Figure R2-1 below): the AI label increases human answer supply; the effect of an AI answer on human answer supply becomes more negative as AI-answer length increases; AI-answer quality does not appreciably shape human answer supply; the AI label does not affect human answer quality; AI-answer length does not affect human answer quality; and the effect of an AI answer on human answer quality becomes more positive as AI-answer quality increases.

We then test these predictions using our empirical data. In light of this revised conceptual framework, we have removed the analysis of viewer recognition and the position effect to keep the paper theoretically focused and internally coherent.

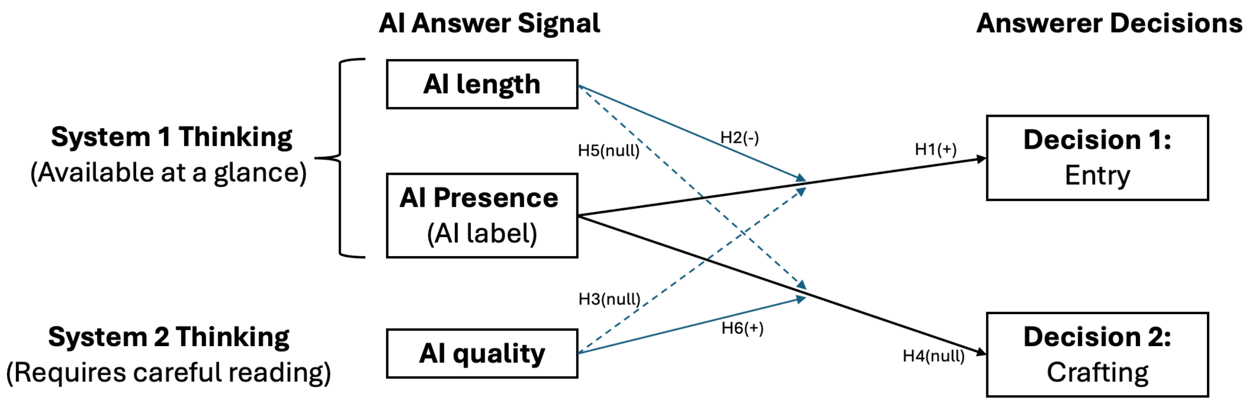

Figure R2-1. Conceptual framework

Under this framework, the **cognitive offloading** argument you suggested is aligned with the two systems of thinking theory**. The crowding out** effort you suggested is aligned with the entry decision, where a comprehensive AI answer deters human answerers (H2). The **anchoring** effect is now the explicit core of H6, where human answerers build on the AI answer to produce their own answers. We appreciate your suggestion to help us improve the paper.

References:

Chen, M.-J. (2009). Competitive dynamics research: An insider's odyssey. *Asia Pacific Journal of Management*, 26(1), 5–25.

Chen, M.-J., Su, K.-H., & Tsai, W. (2007). Competitive tension: The awareness–motivation–capability perspective. *Academy of Management Journal*, 50(1), 101–118.

Evans, J. St. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, 8(3), 223–241.

Kahneman, D. (2003). A perspective on judgment and choice: Mapping bounded rationality. *American Psychologist*, 58(9), 697–720.

Kahneman, D. (2011). Thinking, fast and slow. Farrar, Straus and Giroux.

R2-2

R2-2. In the last round, I asked whether it is theoretically possible that AI answers increase human participation and provide a "feeling that the authors get some empirical findings first and then try to propose some mechanisms to explain the findings."

The authors responded by clarifying that their "theoretical framework was developed before conducting analyses."

Based on the authors' response, "We appreciate your suggestion that AI answers might increase human participation by providing a "starting point." While theoretically possible, this mechanism is not supported by our empirical findings, which provide evidence against such a "starting point" mechanism." the authors admitted the theoretical possibility that AI answers increase human participation. And they choose not to propose it because of their empirical findings.

To summarize, the authors should discuss the theoretical predictions of the positive effect of AI answers on human participation in the front unless they argue that it is not theoretically possible. Using empirical results to guide theoretical propositions is HARKing (Hypothesizing After Results are Known; Kerr, N. L., 1998), as the authors highlighted.

Response

Following your suggestion, we revamp the theoretical framework to discuss AI answers' impact on human participation.

Now, we frame the human decision as two decisions: entry is a System 1 decision because it can be settled quickly before close reading; the crafting decision is a System 2 decision because it is made after the answerer has committed to answering and determines how good the submitted answer will be and how it will be produced. We also differentiate AI into three types of features: label, length, and quality. H1 formally predicts a positive effect of the AI label, which is due to a domain-specific impression that AI is unreliable on genuine technical problems (Tversky & Kahneman, 1973). Moreover, we project that human perception of the AI answer through length (H2) would deter their participation. The overall negative effect is due to the dominance of AI-answer length in our setting.

We appreciate your comments that help us improve. The major changes are in Section 3 Theoretical Development

Reference:

Tversky, A., & Kahneman, D. (1973). Availability: A heuristic for judging frequency and probability. *Cognitive Psychology*, 5(2), 207–232.

R2-3

R2-3. I provided a counterargument for the content effect by arguing that the imperfections of AI can perform worse than humans. The authors argued that "If AI-generated answers are sufficiently comprehensive and well-articulated to satisfy the immediate information needs of potential answerers and questioners, they reduce the perceived gap between existing information and what potential contributors might add, which in turn decreases contribution motivation. Conversely, if AI answers are of insufficient quality, they would not generate this *crowding-out effect. Our empirical findings indicate which scenario prevails in our setting." Additionally, they provide empirical evidence to show that AI answers in their context are sufficiently comprehensive.*

I feel my concerns were not addressed, and the counterarguments hold.

I agree with the authors that if AI performs better than humans, some humans who can not make new contributions may decrease their motivation to contribute. However, the authors didn't argue theoretically that AI performs better than humans in this context; they just said that "If AI-generated answers are sufficiently comprehensive…"

Although they provided some empirical evidence (in the empirical sections) to support that AI performs better than humans, the theoretical arguments about the superiority of AI answers are missing.

Response

We thank you for pressing us on this point. After considering this comment more carefully, we have moved beyond treating the AI answer as an undifferentiated whole or assuming that it must be sufficiently superior to crowd out human contribution. Instead, the revised manuscript conceptualizes each AI-generated answer as a profile of distinct signals. This approach directly incorporates the reviewer's possibility that an AI answer may be imperfect or perform worse than a human answer.

The AI-answer profile consists of the AI label, AI-answer length, and AI-answer quality. The label identifies what kind of competitor produced the answer and may lead contributors in this technical Q&A setting to appraise the AI as a weak competitor, which encourages entry. Length is visible at a glance and signals how much answerable ground the AI appears to have taken, which may reduce the perceived room for another contribution. Quality, by contrast, becomes available only through close reading. Drawing on the two-systems account, we therefore link the glance signals of label and length to the fast entry decision and the reading signal of quality to the more deliberate crafting decision.

This profile approach accommodates different configurations of AI-generated answers. An AI answer may encourage entry because it is perceived as a weak competitor, while its length may simultaneously discourage entry because it appears to cover considerable ground, even when the answer is imperfect. Once a contributor enters, a higher-quality AI answer can provide useful material for crafting a better human answer. We therefore no longer theorize a single content effect that depends on AI superiority.

R2-4

R2-4. I countered the source effect by addressing potential concerns about attributing greater authority or credibility to the AI source, since the authors used the literature on predictive AI to argue for generative AI and avoid discussing algorithm aversion.

In response, the authors argued that algorithm appreciation and aversion are not mutually exclusive, and they posit that "within online technical Q&A communities, the AI source label carries specific credibility signals that influence user behavior." In addition, they argued that there are common contexts between the forecasting contexts (used in Logg et al. 2019) and this research context.

My concerns still hold. The authors didn't address whether the findings in the literature about the authority/credibility in predictive AI in context, such as forecasting contexts (used in Logg et al. 2019), work, and are suitable to argue for the authority and credibility of the generative AI. Even though both predictive and generative AI are part of AI, they work differently. Generative AI used in this research can fabricate and hallucinate. Whether users hold the same level of authority and credibility perception on them as they do on predictive AI is unknown.

Response

We appreciate this comment and have made the following changes in response.

First, we have removed the analogy to predictive AI. We agree that findings on authority and credibility in forecasting contexts, such as those in Logg et al. (2019), may not be transported directly to GenAI, given that predictive and generative AI operate differently and that generative AI can fabricate and hallucinate.

Second, we have adopted the position of AI skepticism that you suggested, and we now ground the AI label effect in the context of our own research setting. In the community we study, at the time our data were generated, AI-generated answers can be unreliable and error-prone on genuine technical problems. This could lead human answerers to view AI as a weak competitor, which in turn motivates their contribution. We have elaborated on this reasoning in our new theoretical framework.

Reference:

Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90–103.

R2-5

R2-5. I provided some counterarguments for the position effect. Specifically, I argued the position effect is negligible because questioners and answerers know the first answer in the treatment group is AI. Therefore, the answerers will work hard to secure the first human position (i.e., the second position in the treatment group). Since both groups of users are randomly assigned, we can assume them to have similar sensitivity to become the first human answerers. Therefore, the position effect is similar across both groups and negligible.

In response, the authors used three points. Position effect is grounded in the literature. Theorizing the position effect does not include the AI source. Empirical evidence shows that AI answers reduce human participation.

My concerns still hold. First, I agree that the position effect exists. Second, I think arguing the position effect has to consider the AI source because the first position in the treatment group is the AI answer, and AI answers work very differently for both the behavior of questioners and answerers from human answers on the platform. According to the authors, the questioners and answerers cannot engage with the AI's answers (e.g., "accept", "like", "comment"). It makes the first position as the AI or the human answers work differently.

Arguing the position effect without accounting for the different mechanisms by which AI and humans work does not make sense.

Third, empirically, AI answers appearing first reduce human participation. This empirical evidence does not mean that answerers tend to contribute less because they can not get the first-position answers. Those mechanisms that the author proposed can explain the reduction effect. In addition, it may be that the AI answers squeeze out simple answers, making those who tend to provide simple answers not contribute.

Response

We thank you for these detailed comments, which helped us clarify the boundaries of our study. Following your suggestions, we have removed the position effect from the manuscript in this revision, so that our updated theoretical framework aligns more closely with both the literature and the phenomenon of interest.

R2-6

R2-6. I provided a different view of the expertise composition shift mechanism. I argued that the experts should remain in the treatment group because they have domain knowledge, enabling them to leverage AI-generated answers to improve their own answers. Additionally, the answerers will tend not to contribute instead of contributing a lower-quality answer when they see an AI answer. Thus, the human answers in the treatment group should not show a decreased viewer recognition.

In response, the authors argued: 1. "The key issue is that expert participation depends on perceived marginal contribution, not merely on their capacity to differentiate." 2. "comprehensive AI answers dramatically reduce the perceived unique value experts can add." 3. "Thus, rational experts will reduce their effort investment in crafting high-quality answers to these questions, leading to the expertise composition shift that we document."

My concerns still hold. First, the authors admitted that expert participation depends on their capacity to differentiate (see the first point in the previous paragraph). Therefore, solely arguing the marginal contribution without arguing the expert's capacity in leveraging AI answers to improve their likelihood to stay and improve their missed answers misses the important pieces. Overall, AI answers can provide a good starting point for experts and save the cognitive costs to improve their answers.

Second, the authors assumed that the AI provides comprehensive answers in their second point. Whether AI can provide comprehensive answers depends on many factors, including whether the platform fine-tuned their internal generative AI, what is their in-built prompt, and whether they tested it using experts as the feedback providers.

Third, my original comment is "In my view, humans are rational, and if they find they can not get recognition for their contribution, they will not contribute (as indicated by the authors, "Anderson et al...") instead of "becoming disinclined to invest substantial effort in crafting high-quality answers." I meant that if humans feel they can not get recognition, they will not provide a human answer, which makes the remaining human answers have a higher likelihood of getting recognition. It is because only those humans who believe they can get recognition after seeing an AI answer. Therefore, I suggest arguing the positive side of the AI answer on *the viewer recognition of human answers.*

Response

We thank you for this thoughtful alternative account. In this revision, we have removed the viewer recognition component from our theoretical model in order to simplify the model and sharpen its focus, consistent with your concern and the SE's framing suggestion. As a result, the theorization and analysis for viewer recognition no longer appear in the paper. We appreciate your suggestions, which have helped us improve the manuscript.

R2-7

Empirical Analysis

R2-7. I raised a concern about the data's validity because the average number of questions per day between 2021 and April 2024 is 65, while the experimental period (Sept. 2023 – Feb. 2024) only has 24 questions per day.

In response, the authors claimed that "This difference does not indicate a data validity issue. Rather, it reflects the natural evolution and variation in platform activity over time." and the "observed activity level during our study period reflects the platform's actual usage patterns at that time."

I am still concerned because the authors did not explain in detail the steps they took to verify their data. Instead of checking the data they have, they should consult with the platform owners. If it is possible, the authors should consult with the platform owners about the potential reasons and ask the platform to share the evolution of the number of questions on the platform across different months. This reason behind the big difference (65 questions versus 24 questions per day) can potentially confound the treatment effects. I bet the platform owners know/want to know the reasons behind as well.

Response

We thank you and have taken concrete steps to be more transparent.

SegmentFault posts all questions publicly, so we were able to reconstruct the full time series of question volume directly from the platform rather than relying only on aggregate averages. Figure R2-7 plots the number of questions and answers per day across the platform's entire history. The plot reveals an initial growth phase followed by a gradual, monotonic decline in activity. This pattern is characteristic of a maturing community: as the repository of existing knowledge grows, users are increasingly likely to find answers to their queries in previous posts, naturally reducing the volume of new questions. This trend reflects the community's organic evolution rather than any anomaly during our study window. The figure of 65 questions per day represents the average over the full multi-year span, whereas the figure of 24 reflects the lower activity level during the later experimental window of September 2023 to February 2024. These two numbers are consistent with a single, long-term trend rather than a data quality problem.

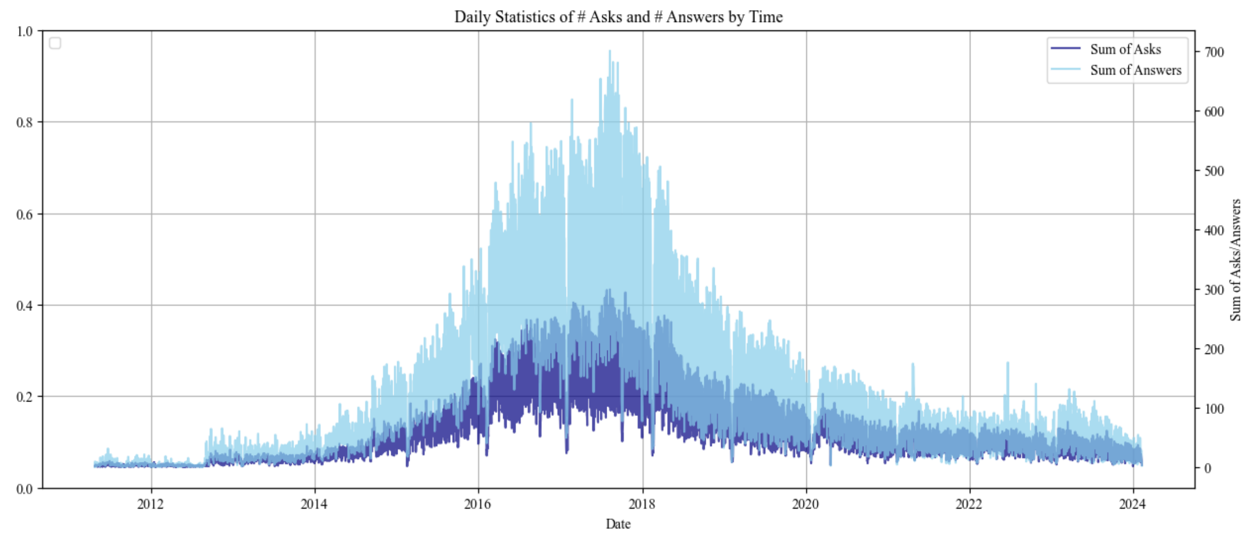

Figure R2-7. The Number of Questions and Answers per Day Across the Platform's Entire History

This trend does not confound our estimated treatment effect. Treatment was randomized at the question level within the experimental window, so any secular change in platform-wide activity applies equally to treatment and control questions and differences it out. In addition, all specifications control for question timing through fixed effects or a cubic polynomial of question age, which absorbs time trends within the window.

R2-8

R2-8. In the last round, I questioned the inconsistency of the sample size. I want to elaborate on my concern in more detail. In the last round of submission, the following text captured my attention.

"From the 3,613 questions in the treatment group, we identify 993 questions with an LLM-enabled content quality rating in the top 25% quantile (scores ranging from 8 to 10), indicating the superior quality of the first answer. Conversely, another 993 questions display quality ratings in the bottom 25% quantile (ratings ranging from 0 to 6), denoting the low quality of the first answer."

There are two inconsistencies. First, there are only 1,766 questions in the treatment group instead of 3,613.

Second, there are 993 questions in the treatment group with the top 25% quantile of their first answer, and another 993 questions in the treatment group with the bottom 25% quantile of their first answer. Only considering those top 25% and bottom 25%, the total number of questions in the treatment group is 1,986, larger than 1,766.

In response, the authors claimed that the total number of questions in the treatment group is 1,766, and the 993 is the matched pairs (993 questions with AI answers matched to 993 questions without AI answers).

I also read carefully about the revised section (section 6.3.2.2), highlighted by the authors. I found my concerns were not addressed. The authors just rephrased the above-quoted paragraph by removing the numbers and avoiding disclosing the number of questions in the *top 25% and bottom 25% subsamples in the analysis. It is concerning.*

For my first concern, it can be a mistake on the authors' side in treating the 3,613 questions as treated questions. But the authors need to respond to the comment and point out how they address it. For my second concern, the authors need to put the real number of treated questions in each subsample.

Response

We apologize for the typos in our previous response.

The treatment group size was a typo in the letter. As reported in Section 4.2, the treatment group contains 1,766 questions; 3,613 is the total across both treatment and control groups. This was purely a reporting error in the letter; all statistical analyses were conducted on the correct samples.

The number "993" was also a reporting error due to a typo. The actual statistical analyses were conducted on correct subsamples, with a high-quality first answer subsample (top 25%): N = 1,697; and with a low-quality first answer subsample (bottom 25%): N = 1,945. Here, we note that this analysis is at the answer level.

Please note that due to the update of the theoretical framework, we removed all reviewer recognition analyses from the manuscript. But we discussed the analysis details here in response to your concern.

R2-9

R2-9. I suggested the authors run a t-test for their randomization checks. The authors followed. Hence, I am good on that.

Response

We are happy you are satisfied with this response.

R2-10

R2-10. I questioned how the authors claimed that AI-generated answers appear within five minutes after question posting. They responded that they monitored/observed the platform in real time.

There are 1,766 questions in the treatment group, which contain an AI answer. Those 1,766 questions are posted across 150 days. I am really curious about how the authors monitor and observe. They should explain the process more clearly. Do they hire students to manually monitor/observe the platforms every five minutes or set up a script to automatically collect the platform every five minutes (does the platform allow them to do so?), or ask the platform to save a snapshot of the question page every five minutes? By saying monitoring/observing in real time without providing the procedures does not help.

Response

We thank you for pressing us on this point. We clarify the procedures below.

Our original understanding of the timing rested on two sources. First, throughout the data collection period, we conducted multiple checks daily to ensure the data collection was proceeding without issues. These daily checks never turned up a treated question that initially lacked an AI answer but displayed one at a later check, which indicates that AI answers did not appear after a long delay.

Second, to examine the timing more closely, we devoted one daytime session to frequent manual observation: each time we opened a newly posted question that contained an AI answer, the question was less than five minutes old and the AI answer was already present. This session, rather than continuous monitoring of all 1,766 questions, was the basis for our original statement.

To obtain a definitive answer about the timing, we consulted SegmentFault's operational manager, who oversaw the experiment. The manager explained the underlying mechanism: the platform assigns each newly posted question to either the treatment or control group, and once a question is assigned to the treatment group, the system sends its content to its LLM and begins answer generation, with no designed delay between the question posting and answer generation.

In light of this, we have removed the general five-minute statement and now state more concisely that "Once the system assigns a newly posted question to the treatment group, it sends the question to an LLM to generate an answer."

R2-11

R2-11. The authors' explanation for the question's zero length makes sense.

Response

We are happy you are satisfied with this response.

R2-12

R2-12. In the earlier round, I suggest that the authors add a question poster/questioner-level fixed effect to remove questioner-level confounders.

In response, the authors clarified that the questioner characteristics are unlikely to confound the effect because treatment assignment is randomized at the question level. Adding questioner fixed effects reduces the sample size substantially, and excluding single-question posters may affect the generalizability. Conducting the question poster random effects and found consistent results.

I would respectfully disagree with the authors' arguments and want to ask for more clarification. First, treatment assignment is randomized, but we still need to test whether the randomization works. The authors run balance tests on observable characteristics and found no systematic differences, but those unobservable characteristics that can confound the results cannot be tested.

Second, adding questioner-fixed effects reduces the sample size substantially. The authors need to show the statistics of the samples in detail. I don't think focusing on questioners *posting on more than one question will lose the generalizability. I may think the platforms want to know whether their way of using generative AI would affect both new and existing users (who post more than one question).*

Third, I appreciate that the authors run the questioner random effect model. However, I may not think the key assumption for random effects models suits this context. To run the random effect model, the authors need to show E(u_i|X_i) = 0, while they can only get E(u_i|treatment) = 0 through the randomization. Therefore, the assumption of running the random effect does not hold.

Response

We thank you for your continued guidance on this point and for encouraging us to examine the identification more closely. We took each of your three concerns seriously and addressed them in turn.

On the sample restriction imposed by questioner fixed effects. Implementing questioner fixed effects requires dropping all posters who contributed only a single question during our observation period, because such observations provide no within-questioner variation. In our data, 934 of the 3,613 questions, that is more than 25% of our sample, were posted by single-question posters. Removing this substantial portion of the data would materially reduce the statistical power of our tests. More importantly, the loss is not merely quantitative but qualitative: the retained sample would consist exclusively of users who post multiple questions, so its results could not speak to single-question posters, who are absent from it altogether. We agree that platforms may want to know whether their deployment of generative AI affects both new and existing users, and it is precisely for this reason that we should favor the random effects specification: it retains the full sample and therefore speaks to both groups. The fixed effects specification, by construction, identifies the effect only for users who posted multiple questions and is silent on single-question posters, a population platforms might also be interested in.

On whether the randomization worked. We first clarify a point that our earlier wording may have obscured: the platform deliberately ran this as a randomized experiment; this is not our own inference from observing that AI answers appeared on some questions. Given this design, we conducted a series of randomization checks on observable characteristics and found no evidence of imbalance, which we continue to report in the revised manuscript. You are correct that unobservable question-level characteristics cannot be directly tested. To address this, we assess whether omitted unobservables could overturn our conclusions using the approach of Oster (2019). The resulting bounds indicate that our estimates are robust to unobserved omitted variable bias. This provides evidence that unobservable confounders are unlikely to drive our results. This also speaks to your preference for a fixed effects specification. A fixed effects specification is valued because it absorbs unobserved, time-invariant user characteristics that might otherwise confound the treatment estimate. Since the Oster analysis already shows that our results are unlikely to be driven by unobservable confounders, the fixed effects specification becomes unnecessary, particularly given the reduction in power it entails.

On the assumption underlying the random effects model. We offer one clarification that we believe resolves the concern. Our parameter of interest is the treatment effect, and the consistency of that coefficient depends only on the error term being uncorrelated with treatment assignment, that is, $E(u_i \mid \text{treatment}) = 0$, which the platform's randomization guarantees by design. The additional covariates enter the model to improve efficiency, not to identify the treatment effect; they are nuisance parameters whose coefficients are not the quantities we seek to estimate. Consequently, the condition required to identify the treatment effect is satisfied by the experimental design itself, and the stronger requirement $E(u_i \mid X_i) = 0$ is not needed to identify the treatment effect.

We hope these clarifications address your concerns, and we are grateful for the guidance that prompted them.

Reference:

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2), 187–204.

R2-13

R2-13. I suggested running question-level aggregation when examining the impact on the answer viewer because the IV and DV are not at the same level. The authors followed and found consistent results. I am good with that.

R2-14. I suggested the author justify why they used the cubic polynomial, and they followed. I am good with that.

R2-15. I suggested the authors verify whether users see the AI-generated answers in the question list page. They followed and provided the Wayback Machine proof. I am good with that.

Response

We are happy you are satisfied with these responses.

R2-14

R2-16. I raised a concern about the results of Table 5 in the last round. Specifically, Table 5 reveals that there is no statistical significance in the differences in the number of answers between the questions before and during the experimental period for the control questions. However, the different numbers disclosed by the authors reveal a big difference. For example, the average number of answers for control questions is 1.372, and the overall number of answers per question (including both the experimental period and pre-experimental period) is 1.9.

The authors responded that the difference between 1.9 and 1.372 is not statistically significant and falls within normal sampling variation.

I respectively disagree. First, the authors need to show how they conduct the statistical analysis. The overall number of answers per question is 1.9, while the experimental period is 1.372, signaling that the pre-experimental period has a number of answers per question higher than 1.9. The authors need to show the numbers and be more transparent.

Second, the authors attribute the difference to "normal sampling variation" without specifying the sampling process and the variance.

Response

We thank you for the comment.

First, we want to clarify that the two numbers 1.372 and 1.9 come from **different populations** and cannot be directly compared to infer a trend. 1.372 is on the control group in the post-experiment period, and 1.9 is on the entire population. Our statement on *"normal sampling variation"* was intended to explain that the two numbers come from different **samples.**

Second, it should be noted that pre-experiment questions have been on the platform longer and have mechanically accumulated more answers over time. For the same reason, a raw comparison confounds time-on-platform with behavioral change. The fact that 1.9 is larger than 1.372 does not necessarily indicate a decrease in the number of answers.

For the spillover test, it is important to distinguish the descriptive measure reported in the summary statistics from the outcome used in the analysis. The value of 1.372 is the mean of # human answers within a full-observation-window reported to describe overall platform activity, whereas the dependent variable in our analysis is the # human answers within seven days. Using this seven-day measure, we compare **pre-experiment questions** (N = 4,137; mean = 1.482, SD = 1.131) against **control-group questions during the experiment** (N = 1,847; mean = 1.305, SD = 1.172). These two groups of questions are essentially the same type of questions without AI answers. Furthermore, we employ a **regression that controls for a cubic polynomial of question age** and other covariates. Once the mechanical age effect is removed, the coefficient on the post-AI deployment indicator is **−0.022 (SE = 0.015;** $p > 0.10$**)**. Our claim that *"the difference is not statistically significant"* referred to this result.

This confirms that the introduction of internal GenAI did not change the patterns of human answering for control-group questions.

R2-15

> *R2-17. In the earlier round, I challenged the authors' way of testing the content quality mechanisms by suggesting that they should first test that the AI answers are of higher quality and second test whether this higher quality AI-generated answers demotivate users from generating answers.*
>
> *In response, the authors followed but argued the validity of their original approach, which only controlled the first answer quality by citing highly influential papers.*
>
> *Since the authors argued, I want to respond and raise the authors' incorrect operations of only including the controls to test a mechanism.*
>
> *First, by including controls, the authors didn't test whether AI answers are of higher quality, leaving their proposed theoretical mechanisms for higher AI answer quality in the air and untested (as supported by the authors' response: "By holding answer quality constant (controlling for it), we remove the portion of the AI effect that works through quality differences.")*
>
> *Second, I agree with the authors that (β_1 - β_1') represents the portion of the effect that operates through answer quality. However, the authors did not test whether this difference was significant in their original submission. Therefore, the authors cannot conclude whether the*

reduced effect of AI answers on human answers stems from answer quality.

Accordingly, in this new approach, we still need to understand whether the effect difference before and after including the answer quality is significant or not (i.e., whether the difference of -0.153 and -0.113 in Table 9 is significant or not).

In addition, the authors only used answer length as the answer quality measure. It can be problematic. The internal GenAI in-built prompt can control the length of AI answers. The authors need to test other quality measures that can also mediate the effects.

Response

We thank you for pressing on both points. Following the review panel and the SE's guidance, we have revised our theoretical model, and these mechanisms no longer appear in the manuscript; we have accordingly removed the associated analyses. We nonetheless address your specific points below for transparency.

On testing the difference between β_1 and β_1'. Testing whether this difference is significant is equivalent to testing the significance of the indirect effect in the mediation analysis framework, which can be done via the Sobel test or its bootstrapped variants common in psychology. We did not run these tests because the economics papers we cite do not employ such tests. There is a principled reason not to. In these designs, x is randomized or plausibly exogenous, but the mediator m in the chain x->m->y is not. Therefore, m is endogenous in the regression of y on m and x and its coefficient has no clean causal interpretation, and neither does any test of the coefficient difference, which is a function of it. For this reason, economists treat the coefficient movement (β_1- β_1') as suggestive, descriptive evidence rather than as an identified causal mediation estimate.

Consistent with this, the applied economics tradition supports mechanism claims not through mediation tests but through design-based evidence: heterogeneous treatment effects that align with the proposed channel, showing the effect where the mechanism should operate and its absence where it should not, and ruling out alternative channels one at a time. The "add m and watch β_1 shrink" regression is one piece of corroborating evidence, not a formal test, so attaching a p-value to the difference would overstate its evidentiary weight.

In short, the coefficient-drop exercise is implicitly an indirect-effect test and could in principle be tested formally. We did not do so because the mediator's coefficient is not causally identified, and because we treated that mechanism evidence as suggestive rather than to make a formal mediation claim that our setting cannot support.

On the measurement of AI-answer quality beyond length. In our revised theoretical model, we theorize answer length and answer quality as separate constructs. We appreciate your suggestions, which have helped us improve the paper.

R2-16

R2-18. I raised concerns about the measurement of the "first-answer seeking tendency." The authors clarified that they measured the preference/tendency using observed choice rather than stated intentions. I think it makes sense and can mitigate my concern.

Response

We are happy you are satisfied with this response.

R2-17

R2-19. I challenged the AI's position effect by arguing that the reduction in the effect of AI answers on the number of human answers stems from the fact that those who tend to generate the first answer usually provide high-quality answers (because they prioritize quality), and then these high-quality human answers deter others from contributing.

In response, the authors conducted an additional analysis by including the hasHighQuality1Ans_i (=1 if the first human answer has an above-median LLM-generated content quality score) as the control variable and found that this variable positively affects the answerNum_i.

I feel my concern was not well addressed. A more direct test is to only focus on the control group and regress the answerNum_i on the hasHighQuality1Ans_i. If it has a positive effect, my early argument was disapproved.

Response

We thank you for proposing this more direct test. In the revised paper, we have removed the position effect due to the update of the theoretical framework. We report the test you suggested for transparency.

We restricted the sample to the control group and regressed *answerNum* on *hasHighQuality1Ans*. The estimated coefficient is statistically insignificant (See Table below). This result does not support the argument that a high-quality first human answer deters subsequent human contributions.

Table R2-17. Testing R2's Alternative Explanation That First-Position High-Quality Answerer Deter Others' Participation

	(1)
	DV: $answerNum_i$
$hasHighQuality1Ans_i$	0.012
	(0.015)
other controls	√
answer timing FE	√
only control sample	√
N	1,451

R2-18

2-20. I commented that the source effect was not tested.

The authors responded that "the source effect is tested in Section 6.3.1.3 Table 11." Specifically, "After controlling for the quality of the first answer (content effect) and whether the first human answerer has a first position seeking tendency (position effect), the estimated *treatment effect becomes smaller but remains statistically significant. This residual effect, the portion of the AI impact that persists after accounting for content quality and position, indicates that the mere fact that an answer is provided by AI (rather than a human) independently affects user behavior.”*

I respectively disagree because there can be other reasons aside from the position and content, like anchoring reference, cognitive offloading, etc., for explaining the residual effect.

Response

We thank you for this point and for the anchoring reference suggestion, which we now incorporate. In the revised framework, we no longer rely on interpreting a residual effect. Instead, we theorize and test the AI label, length, and quality effects as distinct constructs. We have also integrated the anchoring reference into our argument for H6: human answerers build on the AI answer to produce their own answers. Because each mechanism is now derived from theory and tested through a prediction stated in advance, our conclusions no longer depend on attributing the residual effect to any single channel.

R2-19

R2-21. I commented that the experts should be domain-specific. The authors followed. I am good on that.

Response

We are happy you are satisfied with this response.

R2-20

R2-22. I challenged the subsample sizes in the test of the expectation-based mechanism.

The authors responded that they explained in their response to my comment R2-8.

As a clarification to the R2-8, the authors didn’t address my comments regarding this subsample size.

Response

We appreciate your clarification and apologize that our earlier reply to R2-8 did not directly report the subsample sizes for the expectation-based test. To be transparent, the “993 versus 993” figures in that letter were a reporting error. The actual analyses used the high-quality first-answer subsample (top 25%, N = 1,697) and the low-quality first-answer subsample (bottom 25%, N = 1,945). We also note that the revised theoretical framework removes all results on viewer recognition; nonetheless, we report the corrected subsample sizes here for completeness and to confirm that the original analyses were run on the correct samples.

R2-21

Here are some of the new comments I have while reading the revised paper. Since this is a revision, I focused only on comments that raised major concerns.

1. The reputation scores of the questioners are different across the treated and control groups, where the reputation scores of the questioners in the control group are higher. This is concerning. The authors ran the balance tests using the log reputation score, which shows insignificant (p=0.065). How about the raw reputation score?

Response

We ran the balance test on the log reputation score because the raw score is highly right-skewed (see the left panel of the figure below), which violates the normality assumption required for the t-test. The log transformation brings the distribution much closer to normal and better satisfies this requirement (right panel).

To address your specific question about the raw score, we conducted a Mann–Whitney–Wilcoxon test, which requires no normality assumption. The result (W = 1,667,892, p = 0.238) confirms that questioner reputation does not differ significantly between the treatment and control groups.

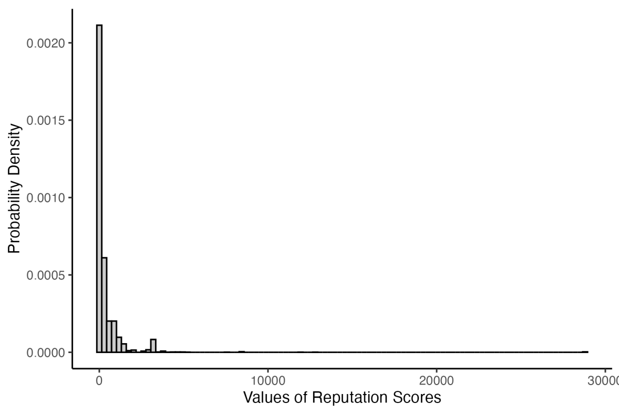

Figure R2-21. Distribution of The Reputation Scores

R2-22

2. The authors progressively add controls that look good, but they need to start with no controls (including removing the question timing FE)

Response

We report the results for a baseline specification with no control variables and no question timing fixed effects. The treatment effect remains consistent in both direction and statistical significance.

We note that our updated theoretical framework has led us to remove all viewer recognition analysis from the manuscript; we nonetheless report these results here for your review.

Table R2-22. The impact of AI-generated answers on the answer supply and viewer recognition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	DV: log of the number of human answers provided within 7 days ($answerNum_i$)				DV: number of net likes received by an answer ($netlikeNum_{ij}$)			
AI_i	-0.052 ***	-0.050 ***	-0.048 ***	-0.046 ***	-0.049 **	-0.052 **	-0.052 **	-0.046 *

	(0.011)	(0.011)	(0.011)	(0.011)	(0.020)	(0.020)	(0.020)	(0.019)
question controls			√	√			√	√
questioner controls				√				√
question timing FE		√	√	√				
answer timing FE						√	√	√
N	3,613	3,613	3,613	3,613	4,670	4,670	4,670	4,670

R2-23

3. The netlike number has negative values. How did the authors adjust the non-negativity for running the Poisson model in column 6 of Table 5? If the authors can shift/adjust netlikeNum for the non-negativity, they can also do it to run a log-transformation model for netlikeNum.

<u>Response</u>

For the Poisson model, we shifted *netlikeNum* by adding a constant "3" (the minimum value of *netlikeNum* is -3) to satisfy the non-negativity requirement.

We also estimated a log-transformation model using log(*netlikeNum* + 3 + 1) as an additional robustness check. The results remain consistent across both specifications and further validate our findings. This analysis is reported below.

We again note that our updated theoretical framework has led us to remove all viewer recognition analysis from the manuscript; we report these results here for your review.

Table R2-23. Robustness checks: alternative model specifications for viewer recognition

	OLS						Poisson
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DV:			$netlikeNum_{ij}$			log [$netlikeNum_{ij}$ +shift + 1]	$netlikeNum_{ij}$ +shifted
AI_i	-0.044 *	-0.044 *	-0.044 *	-0.044 *	-0.046 *	-0.010 **	-0.014 *
	(0.020)	(0.020)	(0.020)	(0.020)	(0.019)	(0.004)	(0.006)
controlling for answer age (age_j)		√	√	√			
controlling for age_j^2			√	√			
controlling for age_j^3				√			
answer timing FE					√	√	√
other controls	√	√	√	√	√	√	√
N	4,670	4,670	4,670	4,670	4,670	4,670	4,670
log likelihood							-7408.622
pseudo R^2							-0.022

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Answer timing FE indicates the date on which the answer was posted.

R2-24

4. Human answerers select to provide answers to questions without an AI answer because they want to gain recognition. The treatment effect is at the answerer level, who can select to join the treatment or control group. Testing the potential spillover effects is the right direction. However, more work is needed. Table 7 should follow Table 3, gradually adding controls, and all robustness checks conducted for the main effects should be applied to test for spillover effects.

Response

We have revised the spillover analysis to mirror the progressive control structure required in R2-22. We also applied all robustness checks conducted for the main effects to the spillover tests.

All results remain consistent across specifications. They show no significant spillover effects. The updated results are reported below.

Table R2-24-1. Testing Potential Spillover Effect

	(1)	(2)	(3)	(4)	(5)	(6)
	DV: log of the number of human answers provided within 7 days (answerNum$_i$)			DV: number of net likes received by an answer (netlikeNum$_{ij}$)		
post-AI deployment control group indicator	-0.027	-0.022	-0.022	-0.024	-0.018	-0.016
	(0.015)	(0.015)	(0.015)	(0.027)	(0.026)	(0.026)
question controls		√	√		√	√
questioner controls			√			√
controlling for a cubic polynomial of question/answer age	√	√	√	√	√	√
N	5,984	5,984	5,984	8,961	8,961	8,961

Table R2-24-2. Robustness Checks For Spillover Effects: Alternative Model Specifications For Human Answer Supply

	OLS		Poisson	ZIP	ZINB
	(1)	(2)	(3)	(4)	(5)
	DV: log of the number of human answers provided within 7 days (*answerNum$_i$*)		DV: the number of human answers provided within 7 days		
post-AI deployment control group indicator	-0.022	-0.026	-0.019	-0.019	-0.019
	(0.015)	(0.015)	(0.042)	(0.042)	(0.042)
other controls	√	√	√	√	√
controlling for a cubic polynomial of question age	√	√	√	√	√
questioner random effect		√			
N	5,984	5,984	5,984	5,984	5,984
AIC		2691.088		17086.635	17086.635
BIC		2979.053			
log likelihood		-1302.544	-8501.317	-8501.317	-8501.317
deviance			5177.952		

pseudo R^2	0.019

Table R2-24-3. Robustness Checks: Alternative Model Specifications For Viewer Recognition

	OLS		Poisson
	(1)	(2)	(3)
DV:	*netlikeNum*$_{ij}$	log [*netlikeNum*$_{ij}$ (shifted) + 1]	*netlikeNum*$_{ij}$ (shifted)
AI_i	-0.016	-0.003	-0.005
	(0.026)	(0.006)	(0.022)
other controls	√	√	√
controlling for a cubic polynomial of answer age	√	√	√
N	8,961	8,961	8,961
log likelihood			-14301.496
pseudo R^2			-0.001

R2-25

5. There is misalignment between the arguments and the empirical test. Arguments: When AI answers appear immediately in the first position, they eliminate this opportunity for humans, reducing potential answerers' incentive to contribute. Empirical test: Our mediator variable, hasFirstPositionSeeker1Ansi, equals one if the first human answer came from such a user. The mediator captures a key prediction of the position effect mechanism: if AI deters participation by occupying the first position, it should particularly discourage "first position seekers" from contributing, making them less likely to provide the first human answer. Not the first human answer, but the overall contribution.

<u>Response</u>

Following your suggestion, we constructed a new mediator variable that counts the number of answerers provided by "first-position seekers" to each question. As shown in Column 2 of the table below, AI significantly reduces this count. When we include this mediator in the main regression, the AI coefficient decreases by a larger magnitude compared to the previous specification, showing that "first-position seekers" significantly reduced their contribution which caused the overall submission reduction.

We note that our updated theoretical framework has led us to remove the position effect from the manuscript; we report these results here for your review.

Table R2-25. Additional Evidence for Position Effect

	(1)	(2)	(3)
DV:	*answerNum*$_i$	*firstPositionSeekerNum*$_i$	*answerNum*$_i$
AI_i	-0.153 ***	-0.261 ***	-0.086 ***

	(0.010)	(0.027)	(0.008)
firstPositionSeekerNum$_i$			0.257 ***
			(0.005)
question timing FE	√	√	√
other controls	√	√	√
N	3,235	3,235	3,235

R2-26

> *6. Likes and comments require different levels of cognitive effort, which is not purely explained by attention-only. There are no significant results associated with commenting activity, but this does not mean there is no attention diversion.*

Response

We agree that likes and comments require different levels of cognitive effort. Our reasoning was that attention is a precondition for commenting: if commenting activity does not change, this is at least suggestive that attention to the questions has not declined. Following the SE's suggestion, we have revamped the theoretical framework and removed the viewer recognition analysis, so likes and comments are no longer in the scope of the paper.

Reviewer 3 Comments

R3-0

I thank the authors for resubmitting the manuscript. I was Reviewer 3 in the previous round. The authors have substantially revised the paper by providing clearer information about the business context, strengthening the empirical analysis, positioning the work more effectively within the literature, and offering additional evidence on the underlying mechanisms. Several of the concerns I raised previously have been addressed, including the validity of the randomized field experiment and the distinction between this study and related work. However, I still have a number of questions that I believe the authors should clarify, as outlined below.

Response

We sincerely thank you for acknowledging our efforts in addressing several concerns. Your continued engagement with our work provides valuable guidance for further improvements. We will carefully address the remaining questions outlined in the review.

R3-1

1. *Mechanism*

In the previous round, I expressed concern that many of the empirical tests did not provide direct evidence in support of the proposed mechanism. I continue to have this concern, as the authors have not substantially revised or extended their analyses in the mechanism exploration part.

a. Content effect

The authors show that the impact of AI on the number of answers is mediated by the length of the first answer, but not by other quality measures generated by the LLM. I do not find this analysis to be strong evidence for a content-based mechanism. First, answer length is not, in itself, a reliable or robust measure of content quality. If the authors are unable to obtain consistent results using other LLM-generated quality measures, does this imply that the content-effect mechanism is not supported, or at least only weakly supported? Alternatively, does this raise questions about the validity and reliability of the LLM-based quality measures themselves? In addition, it would be helpful if the authors could also report results from a formal causal mediation analysis for this mechanism.

Response

We thank you for pressing on the mechanism evidence. On reflection, we agree that answer length is not, in itself, a reliable measure of content quality. In our revised theoretical model, we now treat length and quality as separate constructs: length proxies the coverage that shapes a potential answerer's decision to contribute, whereas quality how good that content is that shapes how good human answers will be. This distinction resolves the tension you identify, since we no longer rely on length as a proxy for quality.

As requested, we also report causal mediation results for the original analysis in Table R3-1, following Imai, Keele, and Tingley (2010). We note, however, that because these mechanisms no longer appear in the revised theoretical model, we do not report these results in the manuscript itself.

Table R3-1. Causal Mediation Analysis Results

	Content Effect	Position Effect
Effect Decomposition		
ACME	**-0.041*** **	**-0.031*** **
	[-0.063, -0.018]	**[-0.037, -0.024]**
Direct Effect (ADE)	-0.115***	-0.125***
	[-0.144, -0.083]	[-0.144, -0.104]
Total Effect	-0.156***	-0.156***
	[-0.175, -0.136]	[-0.176, -0.136]
Proportion Mediated		
% of Total Effect Mediated	**26.50%**	**19.70%**
	[23.6%, 30.4%]	**[17.5%, 22.6%]**

Reference:

Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, 15(4), 309–334.

R3-2

b. Position effect

I also find the analysis of the position effect confusing. The authors examine whether AI reduces the likelihood that the first human answer is provided by a "first-position seeker." However, it is not clear why the focus is specifically on the first human answer. It is also unclear how a user's prior tendency to post the first answer to a question should influence their decision to provide the first human answer (which is, in fact, the second answer overall when an AI answer is present). In addition, even if the presence of an AI answer reduces the number of users who seek to provide the first answer in other questions, this does not necessarily imply that the focal question will receive fewer answers overall; there may be sufficient users who are indifferent to answer position and still choose to contribute. Overall, I am not sure how this analysis provides compelling evidence in support of the proposed position-based mechanism.

<u>Response</u>

We agree with these points and appreciate you articulating them so clearly. Following your suggestion, we have removed the position effect from the manuscript in this revision. Doing so makes the paper more focused and better aligns our theoretical model with the broader literature and the phenomenon of interest.

R3-3

c. Source effect

The authors argue that any residual effect of AI on the number of human answers, after controlling for both content and position effects, can be interpreted as evidence of a source effect. This inference is not well justified for at least two reasons. First, there may be alternative explanations for why the introduction of AI reduces the number of human answers that are not captured by the current controls. Second, any remaining residual effect could simply reflect measurement error or incomplete operationalization of the content and position mechanisms. Indeed, the earlier analysis of the content effect already suggests that the proposed quality measures are not ideal, which further undermines the claim that the residual effect can be cleanly attributed to a source-based mechanism.

Response

We agree with your concerns and have removed this analysis from the paper. In its place, the revised framework no longer relies on interpreting a residual effect. We now posit that the effect of an AI answer depends on a profile of its features, including the AI label, length, and quality. In a community that views AI-generated content with suspicion, users treat this competitor as weak; accordingly, we theorize that the AI label effect motivates human contribution (revised H1). The AI answer's length works in the opposite direction, as a longer answer deters contribution (revised H2). The overall prediction is the joint effect of these forces, which is negative in our sample.

We hope this development, grounded in a coherent overarching framework, is clearer and addresses your concern.

R3-4

d. Attention diversion mechanism

Although the effect of AI on the number of comments is statistically insignificant, the magnitude of the estimated coefficient is very similar to that for net likes. It may be hard to conclude that users' attention has not been diverted from human to AI answers.

Response

We thank you for raising this concern. In light of it, together with the SE's framing suggestion, we have removed the viewer recognition analyses from the manuscript. We appreciate your suggestion, which has helped us improve the paper.

R3-5

e. Quality degradation mechanism

I believe the authors' analysis only shows that users who post answers to questions with AI involvement have a similar level of expertise as those who respond to questions without AI.

This does not, however, rule out the possibility that the same type of users exert less effort on questions where an AI answer is present, which could result in lower-quality human answers.

<u>Response</u>

We thank you for this comment. To address the concern that the same users might exert less effort when an AI answer is present, we regressed the LLM-generated quality ratings of human answers on the AI treatment indicator. As shown in Column 3 of Table 12 in our previous version, AI presence is not associated with human answer quality ($\beta = -0.002$, $p = 0.975$), which is inconsistent with a reduced-effort account. Following your concern and the SE's framing suggestion, we have removed the viewer recognition analyses from the manuscript, but we retain this effort-and-quality analysis within the updated framework to directly address the point you raise.

R3-6

f. Elevated expectation mechanism

I suggest that the authors consider using AI answer quality as a moderating variable to test this mechanism, as doing so may offer a more general and informative assessment of the underlying relationship.

<u>Response</u>

We appreciate this suggestion. Using AI-answer quality directly as a moderator is not feasible in our design: control-group questions never receive an AI answer, so no AI quality measure exists for those observations, which precludes a valid cross-group interaction test. To capture the underlying intuition that higher-quality AI answers should elevate viewer expectations more strongly, we instead use the quality of the first answer as the moderator. As the table below shows, the interaction between AI and first-answer quality is negative and significant ($\beta = -0.037$, $p < 0.05$): when the first answer is of higher quality, AI treatment produces a larger reduction in net likes on subsequent human answers. This directly supports the idea that a high-quality initial answer raises viewers' evaluative standards, leading to harsher assessments of later human contributions.

We note that, following your broader concern and the SE's framing suggestion, we have removed the viewer recognition analyses from the manuscript. We nonetheless report this result here and appreciate your suggestion, which has helped us improve the paper.

Table R3-6. Additional Test for Elevated Expectation Mechanism

	(1)
	DV: number of net likes received by an answer ($netlikeNum_{ij}$)
AI_i	0.169
	(0.103)
AI_i*$quality1Ans_{ij}$	-0.037 *
	(0.015)
other controls	√

answer timing FE	√
N	3,189

R3-7

2. Theory

In Section 3, the authors seek to theorize how internal GenAI may influence users' contributions to answers. However, there does not appear to be an overarching theoretical framework that explains why all three effects should shape contribution decisions. At present, each effect is discussed in isolation, which makes the overall theoretical argument appear somewhat ad hoc.

Response

Following your suggestion and the guidance provided by SE, we have developed a new theoretical framework centered on the competitive encounter between human answerers and AI answers generated by the platform's internal GenAI system. Specifically, we theorize how the AI answer affects two decisions made by human answerers: the **entry decision**, or whether to contribute an answer at all, and the **crafting decision**, which concerns how good the human answer will be and how it will be produced, if the answerer decides to contribute.

We adopt a competition lens informed by competitive dynamics theory (Chen et al., 2007; Chen, 2009). This lens matches the setting because the AI answer sits in the same thread and on the same screen as any human answer that follows. A later human answer must therefore be worth reading in the presence of the AI answer. It also gives us richer predictions than a simple substitution account, which would predict only that AI answers reduce human contributions. By contrast, the competition lens explains why an AI answer may discourage human entry in some cases while eliciting higher-quality human answers from those who do enter.

The revised framework treats the AI answer as a profile of three signals:

(a) **AI label.** The label identifies the source of the existing answer and tells the answerer what kind of competitor this is. Because machine authorship is apparent at a glance, the label can shape the answerer's initial appraisal before the AI answer's content is read. In our theory, the label leads answerers in this technical Q&A setting to appraise the AI as a weak competitor, which strengthens the incentive to enter.

(b) **AI-answer length.** Length is also available at a glance. It signals how much of the answerable ground the AI answer appears to have taken. A longer AI answer can appear to cover more possible causes or fixes, which reduces the perceived room left for a human contribution and weakens the incentive to enter.

(c) **AI-answer quality.** Quality is not available at a glance. It becomes available only when the answerer reads the AI answer and evaluates whether its reasoning, diagnosis, or code fits the specific problem. Once a human answerer has decided to contribute, a higher-quality AI answer serves as a better foundation to build on, either by supplying adaptable material such as a correct diagnosis or working code that can be reused, or by charting a viable direction that the answerer can develop into a full answer.

We link these signals to the two decisions through the two systems of thinking theory (Evans & Stanovich, 2013; Kahneman, 2003, 2011). The AI label and AI-answer length are **glance signals** processed through fast, low-effort System 1 thinking, whereas AI-answer quality is a **reading signal** processed through deliberate System 2 thinking. Similarly, entry is a System 1 decision because it can be settled quickly before close reading, the crafting decision is a System 2 decision because it is made after the answerer has committed to answering and determines both how good the submitted answer will be and how it will be produced.

This classification yields a **same-level principle**: a signal affects a decision when both occupy the same level of processing, while cross-level effects are expected to be null. Thus, the AI label and AI-answer length shape entry, but AI-answer quality does not. Conversely, AI-answer quality shapes human answer quality by serving as a foundation the answerer can reuse and redirect to lower the cost of producing a high-quality answer, but the AI label and AI-answer length do not systematically shape human answer quality once the answerer has entered.

In line with this framework, the revised manuscript develops six hypotheses (see Figure R3-7 below): the AI label increases human answer supply; the effect of an AI answer on human answer supply becomes more negative as AI-answer length increases; AI-answer quality does not appreciably shape human answer supply; the AI label does not affect human answer quality; AI-answer length does not affect human answer quality; and the effect of an AI answer on human answer quality becomes more positive as AI-answer quality increases.

We then test these predictions using our empirical data. In light of this revised conceptual framework, we have removed the analysis of viewer recognition and the position effect to keep the paper theoretically focused and internally coherent.

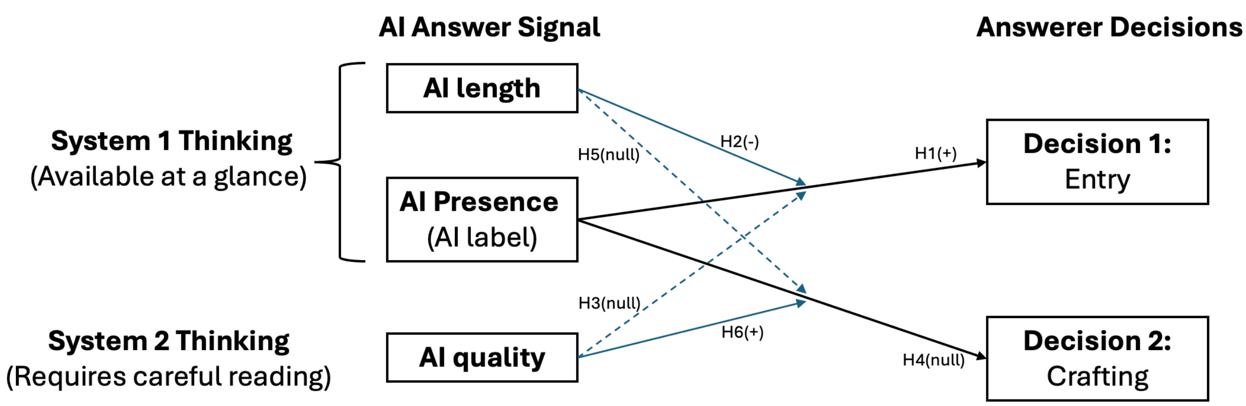

Figure R3-7. Conceptual framework

We have revised the paper according to the revamped framework. We appreciate your suggestions on this.

References:

Chen, M.-J. (2009). Competitive dynamics research: An insider's odyssey. *Asia Pacific Journal of Management*, 26(1), 5–25.

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Evans, J. St. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, 8(3), 223–241.

Kahneman, D. (2003). A perspective on judgment and choice: Mapping bounded rationality. *American Psychologist*, 58(9), 697–720.

Kahneman, D. (2011). Thinking, fast and slow. Farrar, Straus and Giroux.

R3-8

> *3. Other minor issues*
>
> *a. On page 18, the authors state that "we observe that AI answers appear within five minutes of question posting." Could the authors verify whether AI answers consistently arrive faster than human answers?*

Response

Thank you for raising this issue. To obtain a definitive account of the AI answer timing, we consulted SegmentFault's operational manager, who oversaw the experiment. The manager explained the underlying mechanism: the platform assigns each newly posted question to either the treatment or control group, and once a question is assigned to the treatment group, the system sends its content to its LLM and begins answer generation, with no designed delay between question posting and answer generation.

Therefore, by design, this mechanism ensures that AI answers consistently arrive faster than human answers. This is also consistent with our own daily checks and manual observation during data collection.

R3-9

> *b. The authors state that they collected all questions and answers posted between September 2023 and February 2024. However, the analyses presented later in the paper also make use of pre-treatment data. The authors may want to correct this statement.*

Response

We apologize for the confusion. Our main experimental analyses focus on the 150-day window from September 2023 to February 2024. Certain analyses, however, such as the spillover analysis comparing user behavior before and after the experiment began, also draw on pre-treatment data to establish baseline patterns. We have revised the description in the data collection section to accurately reflect both sources of data used in the paper.

R3-10

> *c. What is the distribution of the variable netlikenum? Does it exhibit a skewed distribution?*

Response

Thank you for this question. The distribution of *netlikeNum* is indeed highly skewed (see the figure below), with a mean of 0.288 and a standard deviation of 0.671.

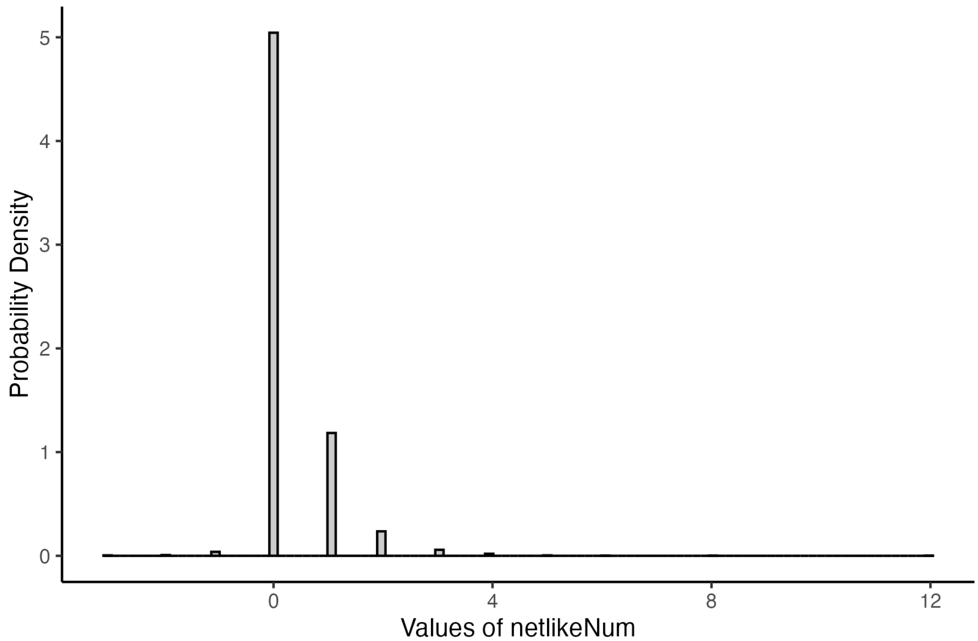

Figure R3-10. Distribution of the netlikeNum

To address this skewness, we re-estimated the model using both a Poisson specification and a log-transformation specification. The results are consistent across all specifications, as shown in the table below.

We note that, to keep the paper focused and make updated theoretical framework aligns more closely with both the literature and the phenomenon of interest, we have removed the viewer recognition analyses from the manuscript. We nonetheless report this analysis here for your reference.

Table R3-10. Robustness checks: alternative model specifications for viewer recognition

	OLS						Poisson
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DV:			$netlikeNum_{ij}$			log [$netlikeNum_{ij}$ +shift + 1]	$netlikeNum_{ij}$ +shifted
AI_i	-0.044 *	-0.044 *	-0.044 *	-0.044 *	-0.046 *	-0.010 **	-0.014 *
	(0.020)	(0.020)	(0.020)	(0.020)	(0.019)	(0.004)	(0.006)
controlling for answer age (age_j)		√	√	√			
controlling for age_j^2			√	√			
controlling for age_j^3				√			
answer timing FE					√	√	√
other controls	√	√	√	√	√	√	√

N	4,670	4,670	4,670	4,670	4,670	4,670	4,670
log likelihood							-7408.622
pseudo R^2							-0.022

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Answer timing FE indicates the date on which the answer was posted.

R3-11

> *d. In the analysis presented in Section 7.2, should the authors account for the content similarity between the AI-generated answer and all human-generated answers?*

Response

Thank you for this suggestion. The analysis in the original Section 7.2 tests whether human answerers adjust their contributions to differentiate themselves from AI content. We deliberately compare only the first and second answers to avoid the confounding influence of other prior answers. Measuring the similarity between the AI answer and all human answers, as you suggest, would instead capture how much human answers differ from AI content overall; because it lacks a control group, it would not isolate whether humans strategically adjust their answers in response to a prior AI answer. We hope this clarifies our design choice.